

Recent SLS Rocket Applications using Overset Grids and OVERFLOW

Overset Symposium
September 21-24, 2026

Guy Schauerhamer
CFDRC / NASA Ames

Stuart Rogers
NASA Ames, Retired

Derek Dalle, Jamie Meeroff, Aaron Burkhead, Josh Diaz
NASA Ames

Purpose

- Overview of Ames SLS CFD Team's efforts with overset grids and OVERFLOW
- Show results from latest SLS configurations

Overview

- Artemis II mission
- Overset grids
- OVERFLOW Setup
- Results of recent SLS configurations



Image credit: NASA

Artemis II









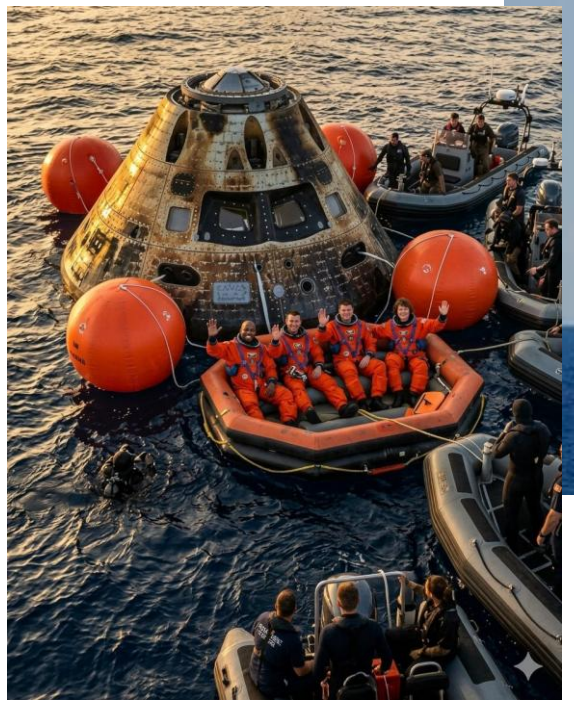




Image credit: NASA

Crew on Zero Deck

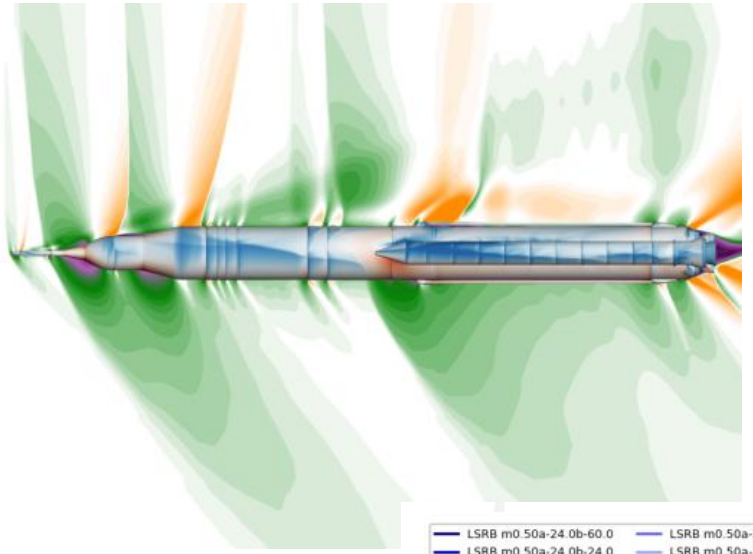
Crew took a moment to look at the vehicle and shake each others' hands



Image credit: NASA

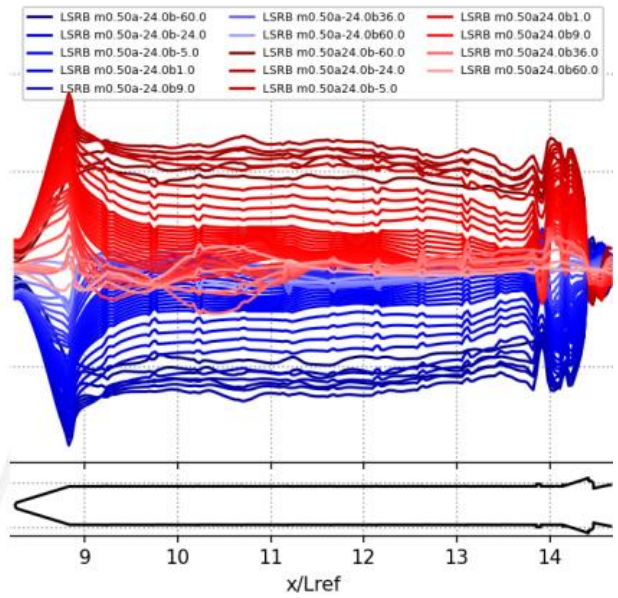
NASA Ames SLS CFD Team

- Current: Derek Dalle, Jamie Meeroff, Aaron Burkhead, Josh Diaz, Guy Schauerhamer
- Former: Stuart Rogers, Henry Lee, Albert Le, Jeff Housman, Cetin Kiris, William Chan, many others



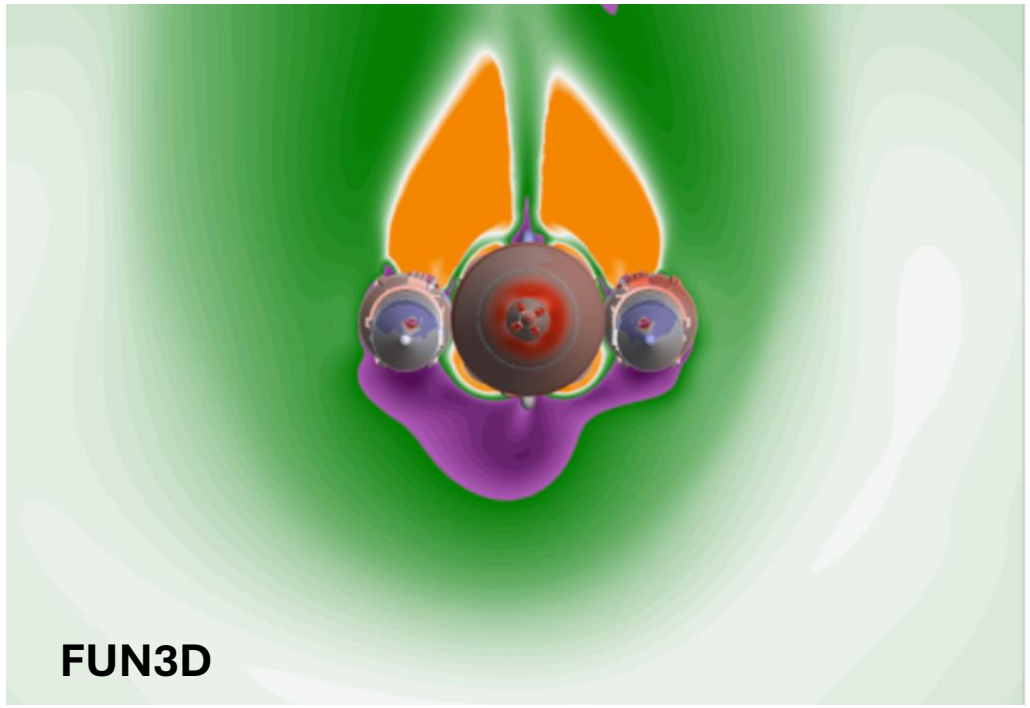
What the team does

- Aerodynamic databases for ascent and booster separation
  - Protuberance aero loads
  - Line loads
  - Surface pressures
- Day of Launch analyses and console support
- Debris Transport Analysis
- Software development
  - **CAPE**- used for case management and database generation for many solvers (Cart3D, OVERFLOW, FUN3D, Kestrel, Loci-Chem, LAVA Cartesian)
  - **LFC**- large files in git repos
  - **Pycnic**- automated CFD simulation convergence evaluation using machine learning
  - **NASA-GPT**- AI Chatbot that scraped NASA technical reports and papers
  - **CAPE-agentic**- CAPE with AI
    - Example command: “check my cases for DONE status and make those reports”
  - **Pegasus5**- not part of SLS but used by SLS team
  - **DTA tools**- collaboration with JSC



CFD Database Generation

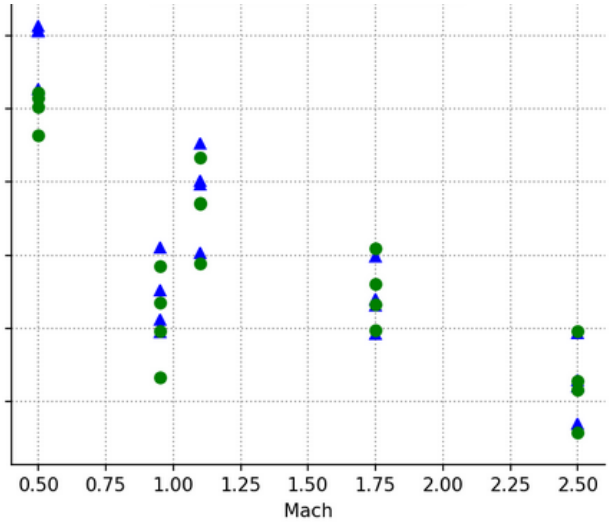
- The CFD databases are completed with FUN3D
- Code-to-code uncertainty quantification completed with OVERFLOW
- Differences between the codes:
  - Structured overset vs. unstructured meshes
  - Structured overset grid is more refined than the unstructured grid
  - Different numerical methods, same turbulence model
  - Different AMR methodologies
  - OVERFLOW multi-gamma plume modeling vs. FUN3D's air exhaust



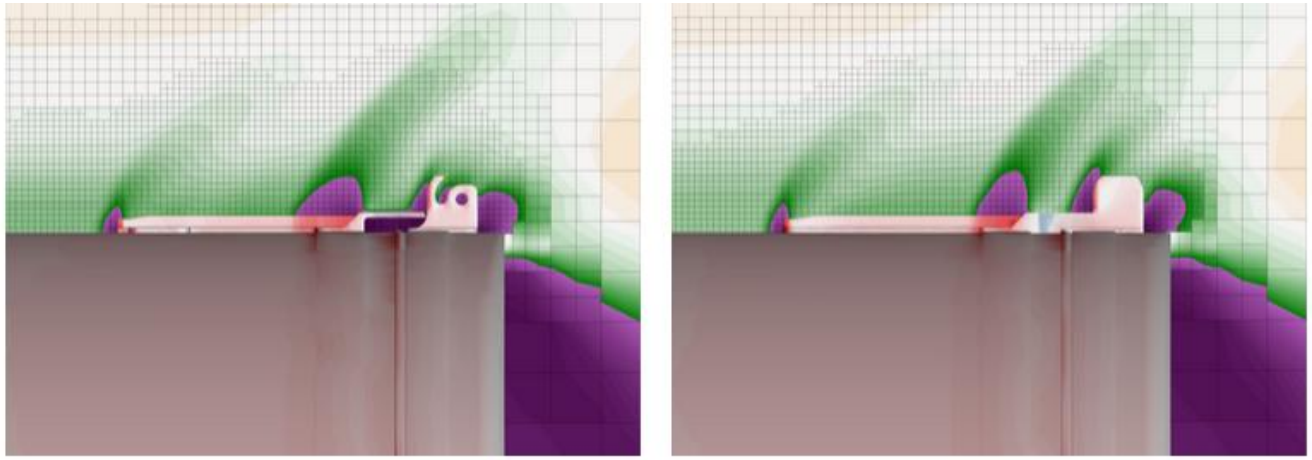


Geometric Simplifications

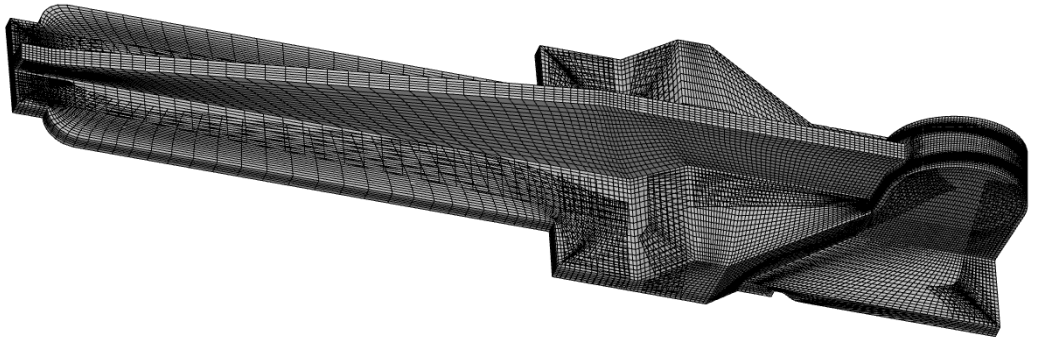
- Use Cart3D to test different geometric detail levels for each protuberance
  - Data driven decisions
  - Less engineering judgement or trying to decrease meshing difficulty
  - Inviscid limitations
- **H** level is full detail, **P01-Pn** decrease detail
- Run Cart3D cases and compare results
- Can’t simplify too much, being used for aero loads databases



SM Panel Hinge



H01 P01



CGT Grid System Methodology

- CGT Build scripting system
  - BuildSrf, BuildVol, BuildPeg5i, BuildMixsuri, BuildOveri
  - GlobalDef.tcl define the configuration
  - Major configurations have individual repos
- Each part lives in its own directory with its own local inputs
  - Many protuberances are reused from previous configurations
- BuildBodyCurves.tcl library
  - Written by Stuart Rogers and Henry Lee
  - Used for axisymmetric bodies and for most raceways
  - Input set of curves, define fomoco breaks with either grid numbers or streamwise coordinate
  - Redistributes, concatenates, and outputs curves for each fomoco component

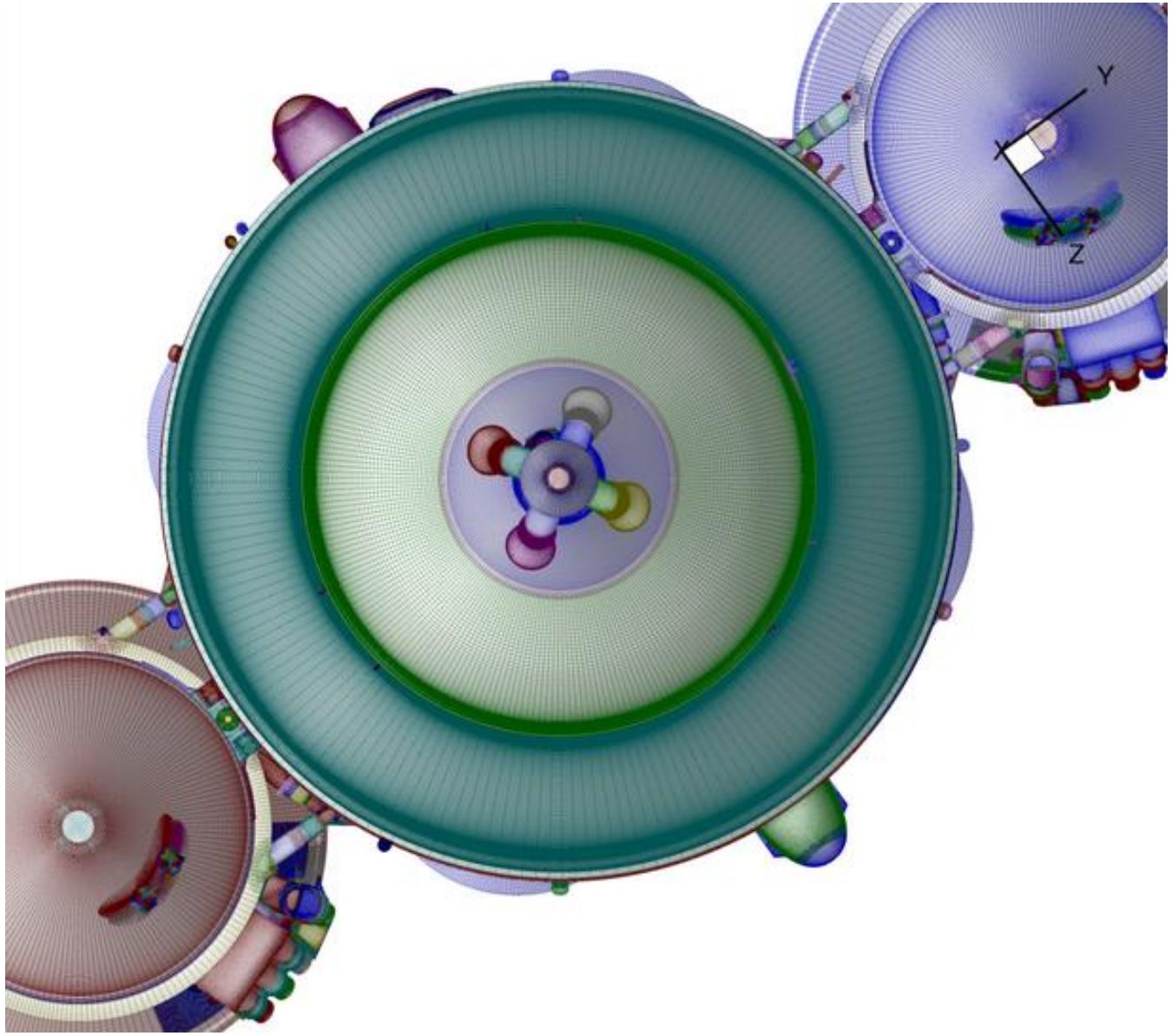
Launch Abort System curves



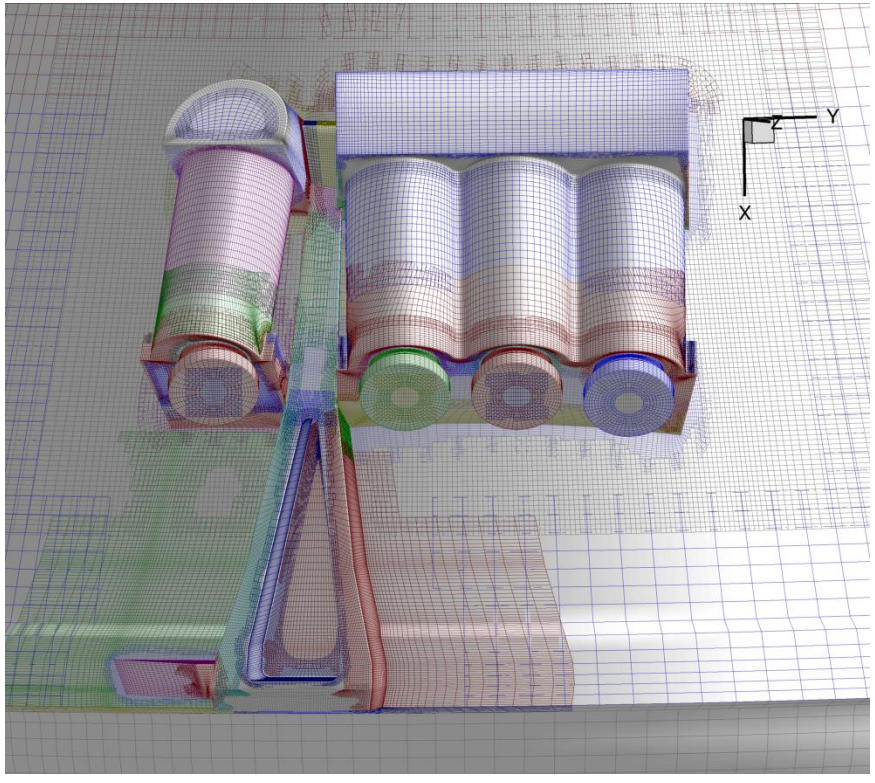
```
├── GlobalDefs.tcl
├── GlobalScaleFactor.tcl
├── config.tcl
├── inputs.tcl
├── CMA_Camera
├── CS_
├── CS_EngineNoz_Flight
├── CS_ESCamera
├── CS_GH2PressLine
├── CS_GOXPressLine
├── CS_GOXVentHood
├── CS_ITCamera
├── CS_ITStrake
├── CS_LOXFeedLine
├── CSM_Umbilical
├── CS_PressLineBracketA
├── CS_PressLineBracketB
├── CS_PressLineBracketC
├── CS_PressLineBracketD
├── CS_PressLineBracketE
├── CS_PressLineBracketF
├── CS_PressLineBracketG
├── CS_PressLineBracketH
├── CS_PressLineBracketI
├── CS_PressLineBracketJ
├── CS_VSSBracket
├── ICPS_
├── LAS_AbortNoz
├── LAS_TowerTunnel
├── LVSA_
├── ORION_
├── RSRB_AftBSM
├── RSRB_BodyIEA
├── RSRB_Flight
├── RSRB_FwdBSM
├── RSRB_HoldDown
├── RSRB_MainTunnel
├── RSRB_SecTun
├── RSRB_TVBrace
├── RSRB_TVCExhaust
├── SM_PanelHinge
├── STACK_AftAttach
├── STACK_FwdAttach
└── STACK_Xrays
```

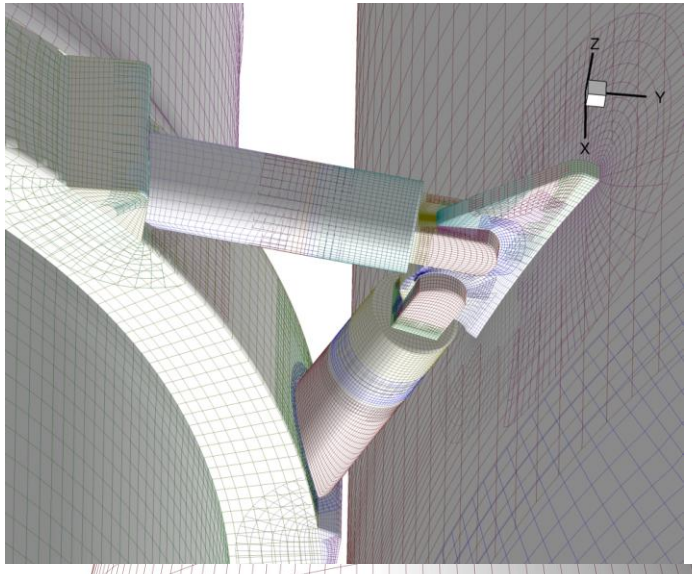
SLS Overset Grid Scripts Contributors

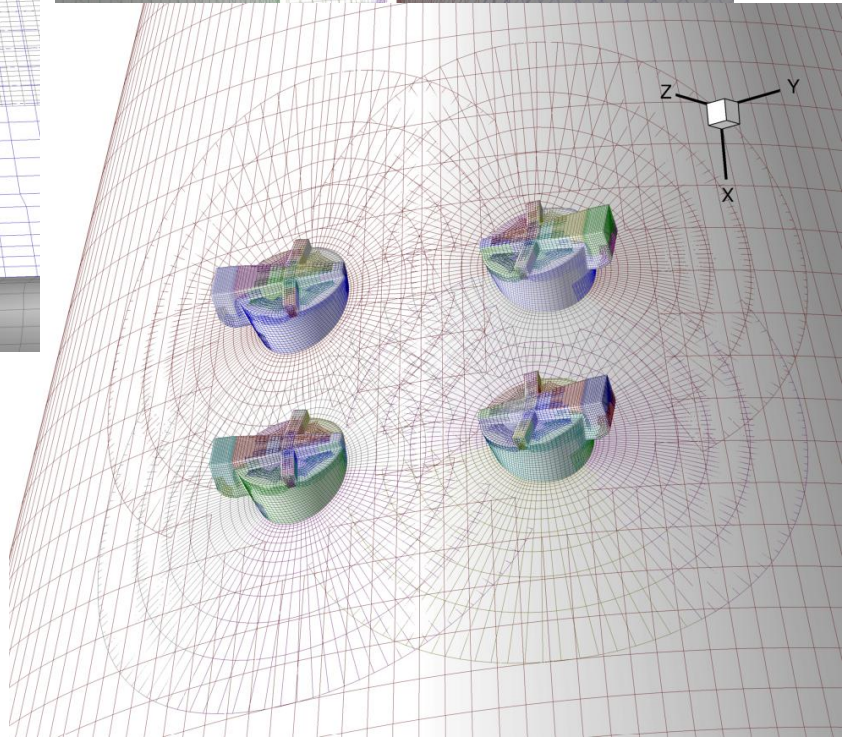
- The current CGT SLS grid scripts have code written by:
  - Stuart Rogers
  - Guy Schauerhamer
  - Henry Lee
  - Derek Dalle
  - Ryan Rocha
  - William Chan
  - Josh Diaz

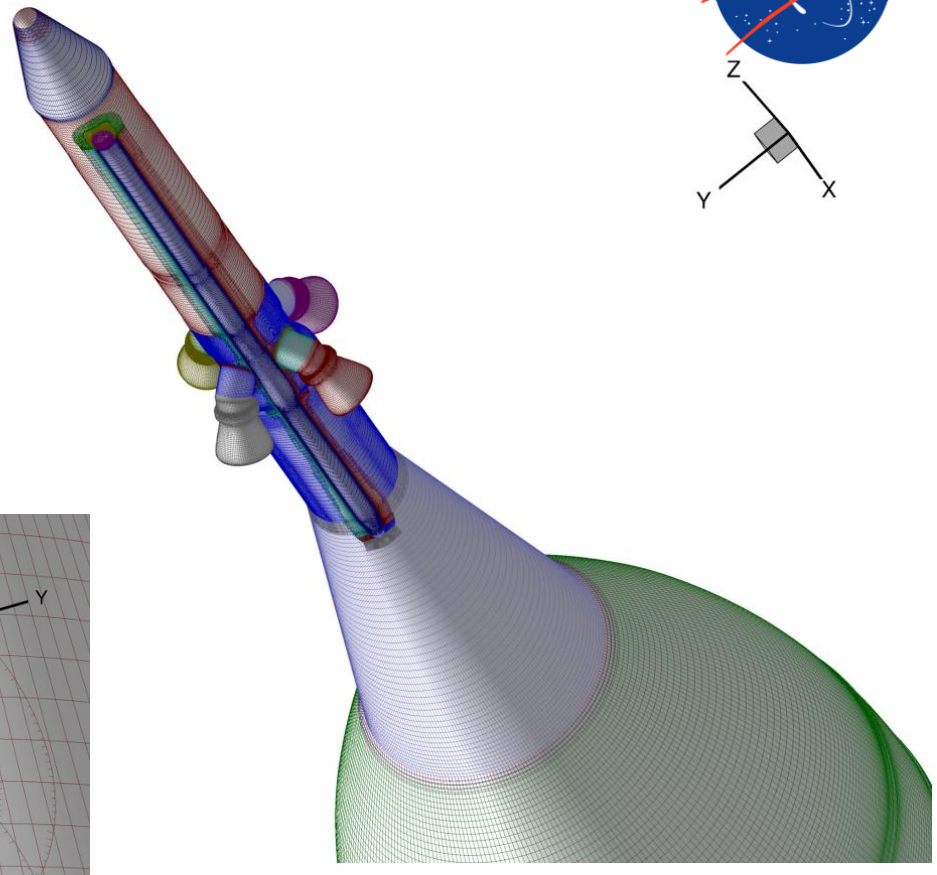


Example mesh images

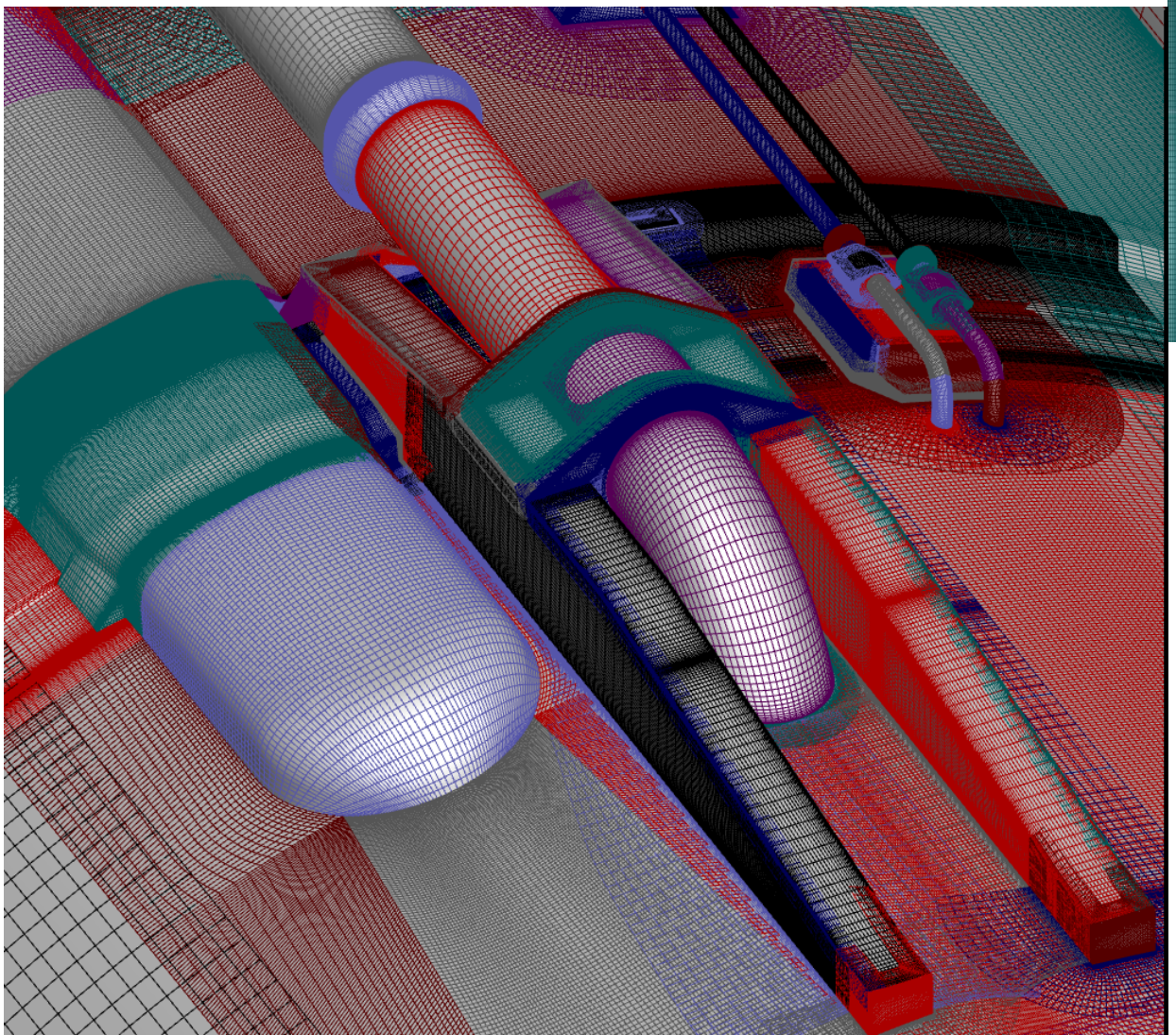


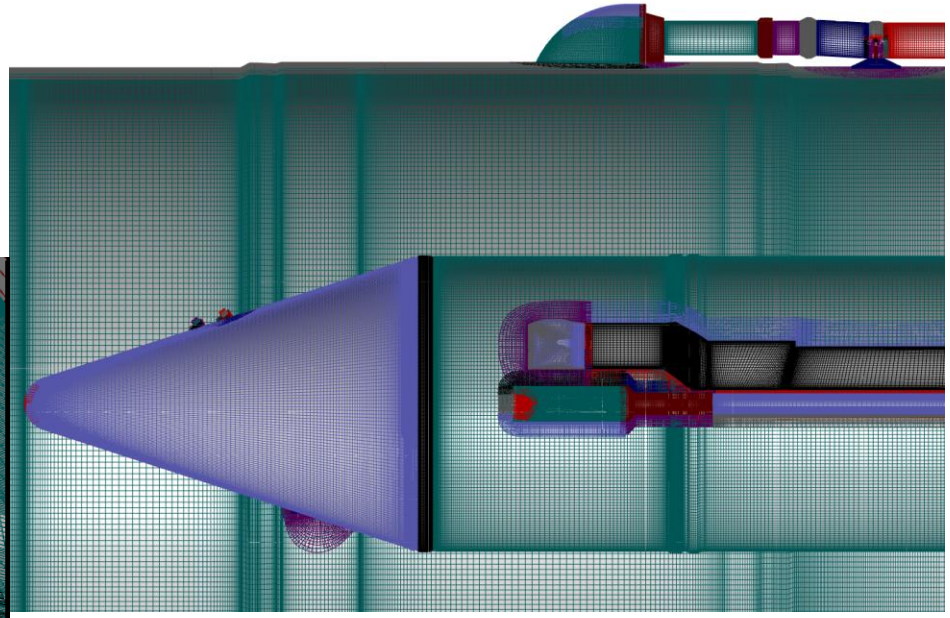


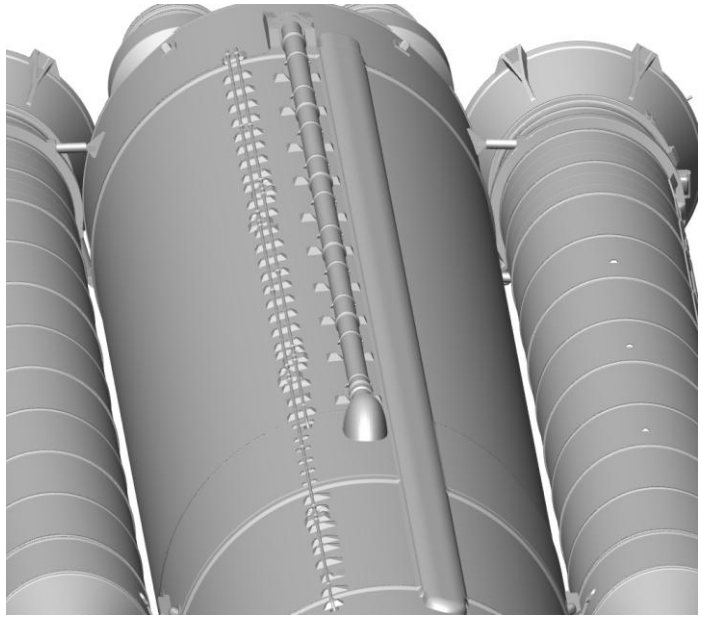




Example mesh images

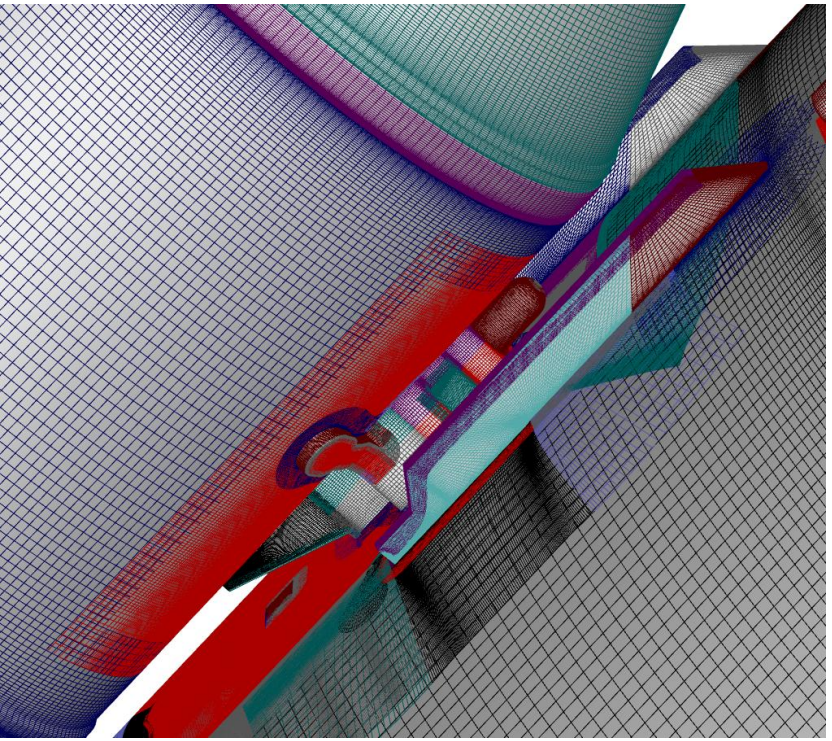


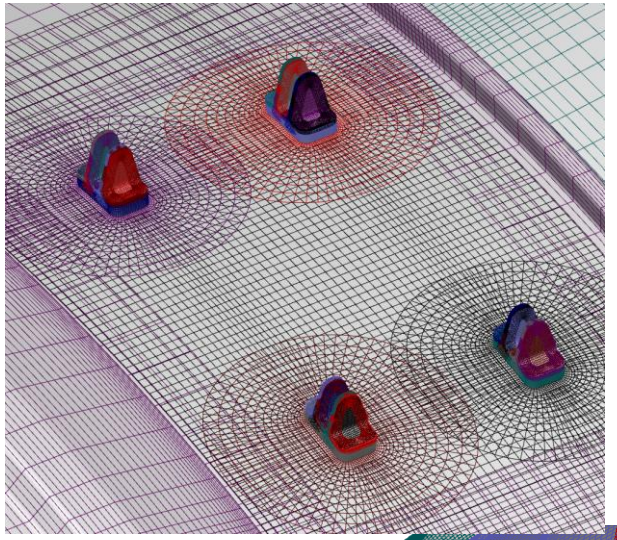


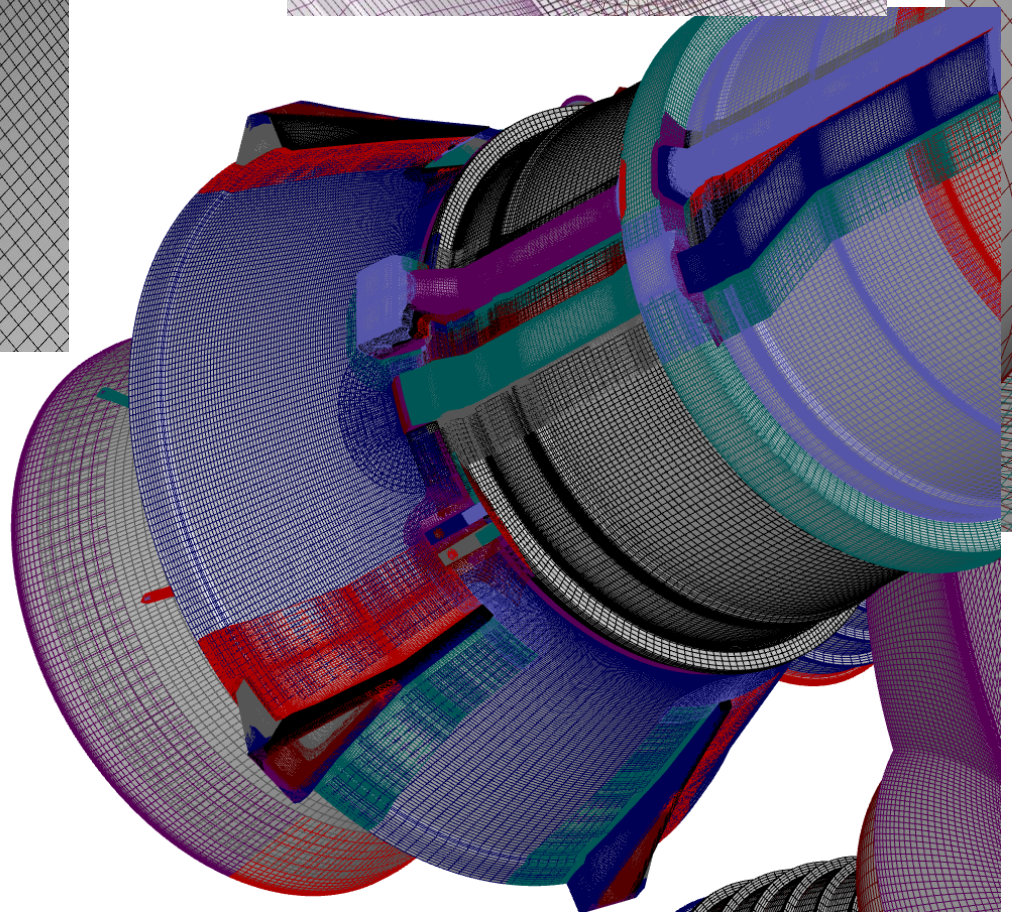


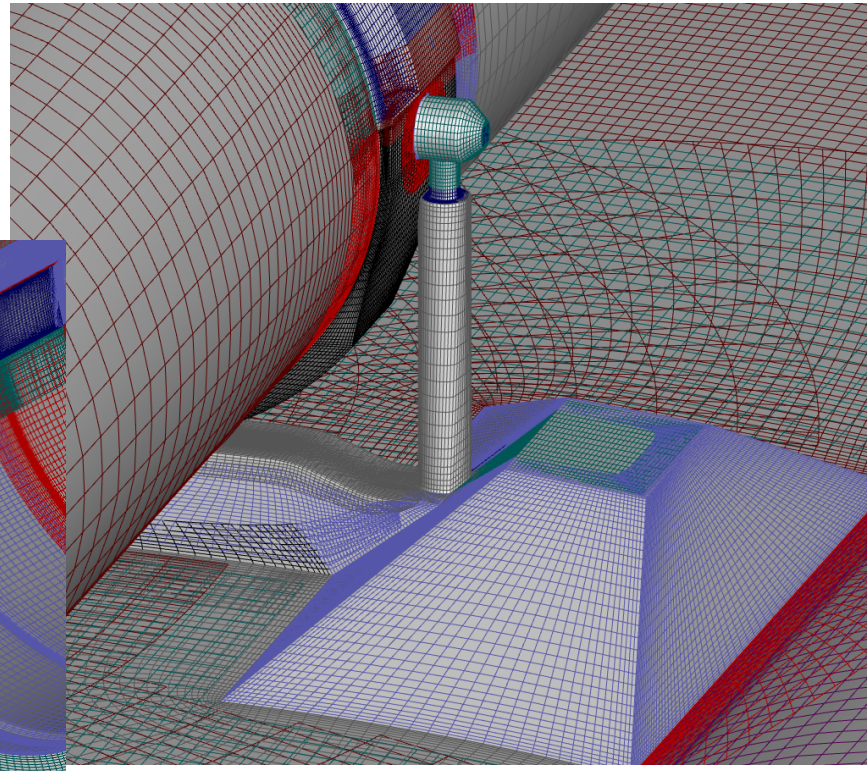
20 LOX line brackets
46 Pressure line brackets
    10 different types
    1 bracket intersection

Example mesh images









Overset Grid System Example



- Many authors means many methods used to build different components
- Start from uh3d definition, convert to triangles, seamcrt (CGT) to get definition curves
- Some scripts are more re-usable than others for geometry updates
  - Automatically sort input curves vs. hard-code input curve numbers
- One example of a systematic approach to meshing protuberances:

```
#*************************************************************************
proc CreateCSM_Umbilical {} {

    global Par Fomo Ovr default env nfomo

    puts "Building CSM_Umbilical"

    # Going to take advantage of symmetry and build just half then mirror

    # Curves
    PrepCurves crvs
    RedistributeCurves t.1

    # Surfaces
    SurfaceDefs t.2
    OversetSurfaces t.3
    Half2Full t.4

    # Position
    PositionSurfs t.5

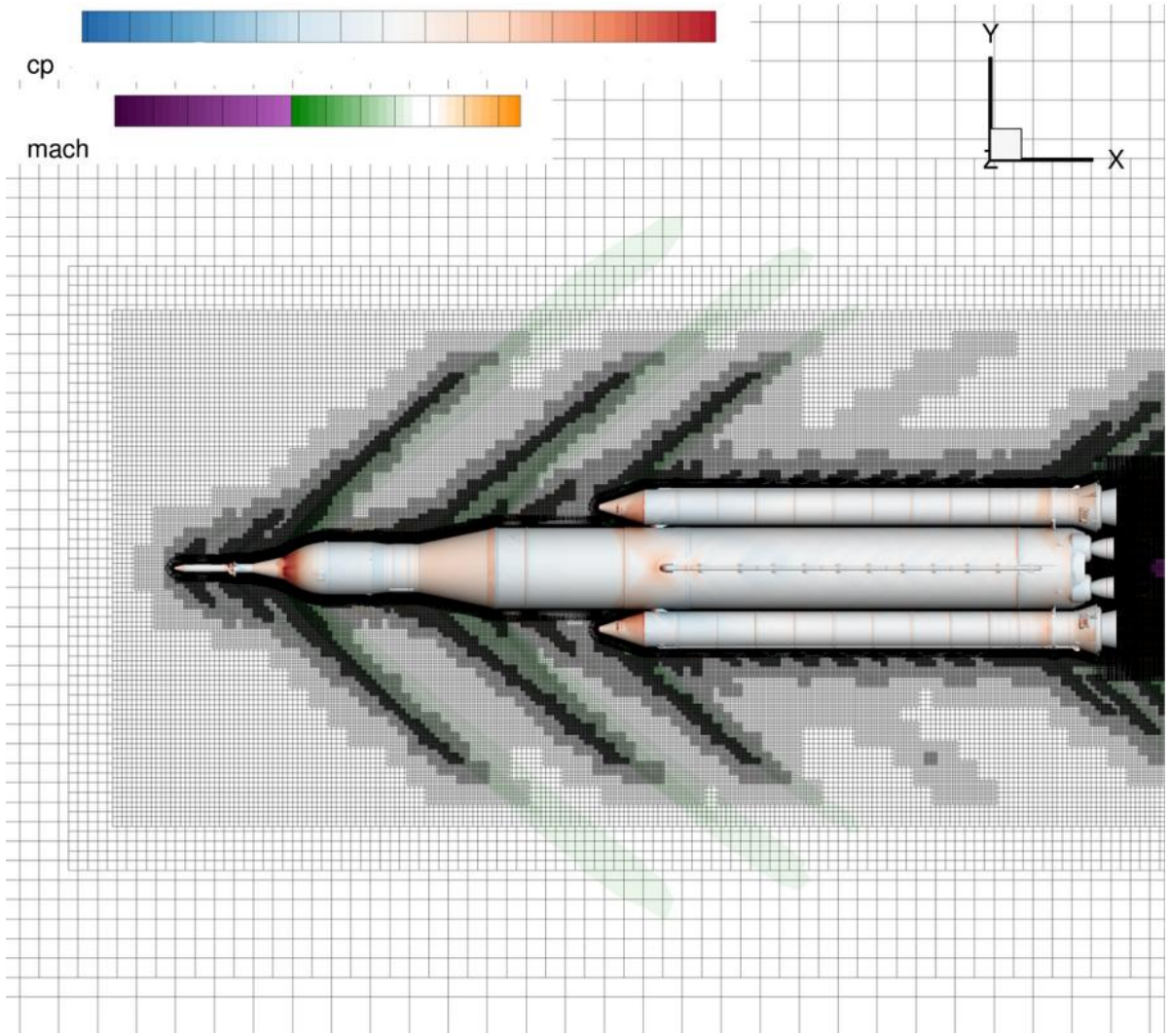
    # Ready for pegasus5
    SrfsandOvfis

    # Clean up
    RemoveFiles t 1 10
    exec /bin/rm -f surgrd.i refs.com crvs

}
```

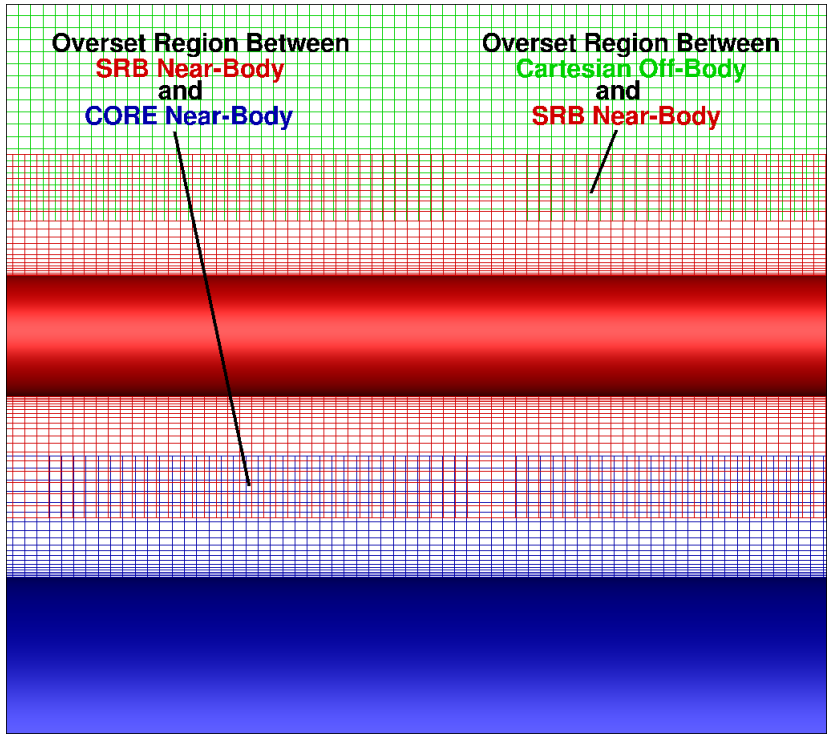
OVERFLOW Setup

- Employed hybrid Pegasus5/DCF domain connectivity
  - Pegasus5 with optimized overset regions in near-body grids
  - DCF to cut Cartesian off-body grids away from CORE and SRB surfaces
- Off-body auto mesh generation and AMR
- HLLC, SSOR, SA turbulence model
- Constant CFL with ramping
- GLS for some configurations
- Start with FMG
- Time accurate with Newton subiterations when necessary to obtain steady state solution

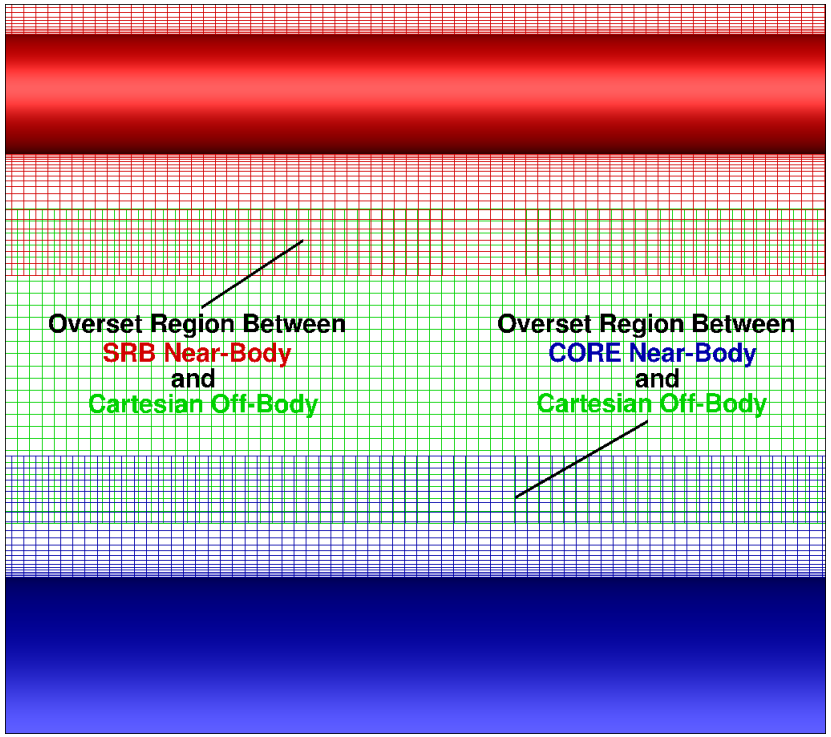


Booster Separation Setup

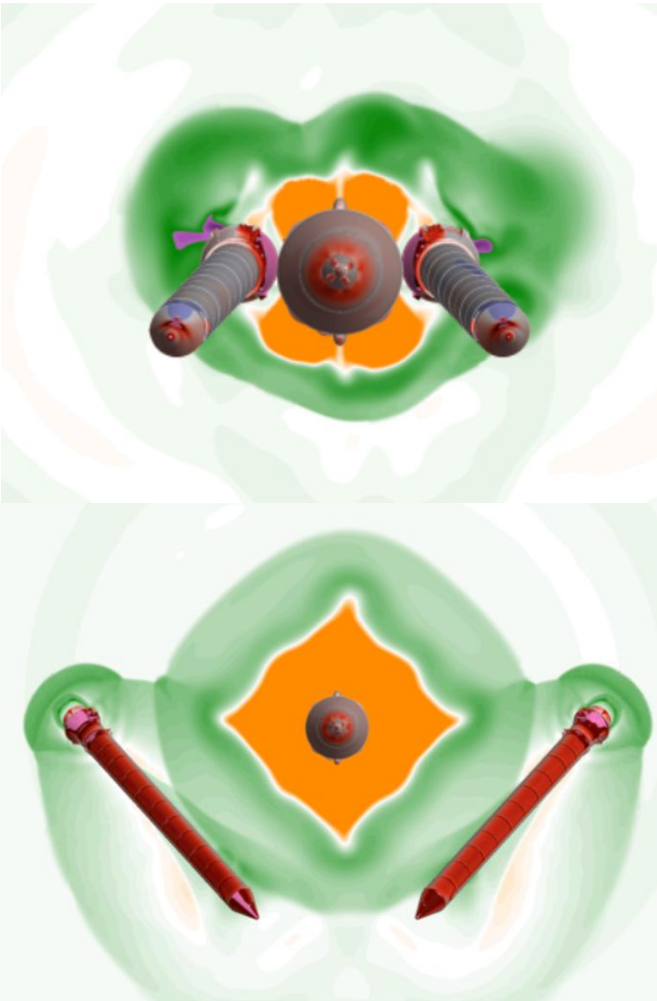
- When SRBs and/or CORE were close, use Peg5 to optimize overset regions between SRBs and/or CORE
- When SRBs were far from each other and CORE, use GMP and x-rays
- Process automated with Python script to complete run matrices
  - Detects volume grid overlap and collisions



Use Pegasus5 between SRB and CORE



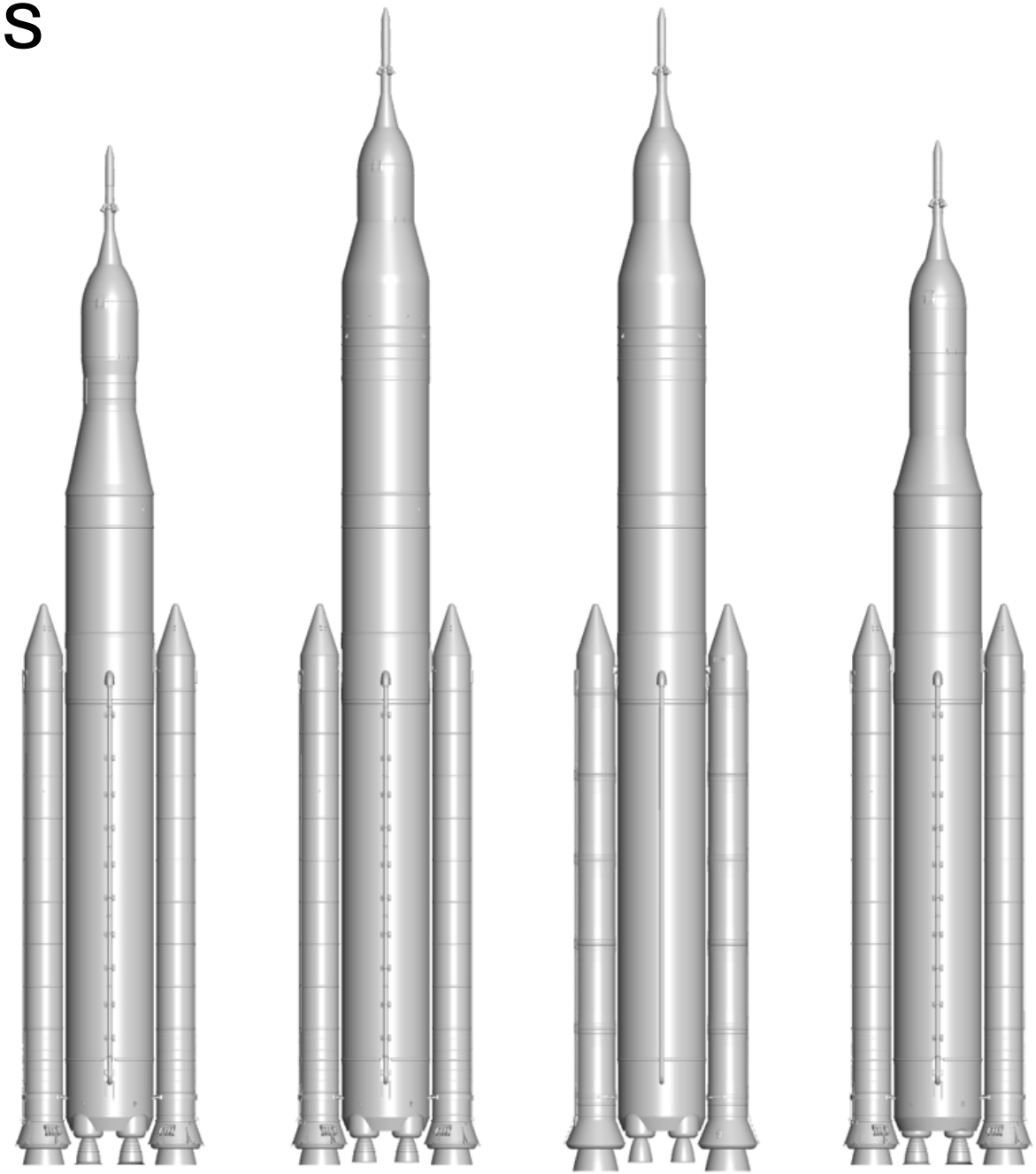
Use GMP and x-rays



Recent SLS Configurations

Modeled with overset grids and OVERFLOW

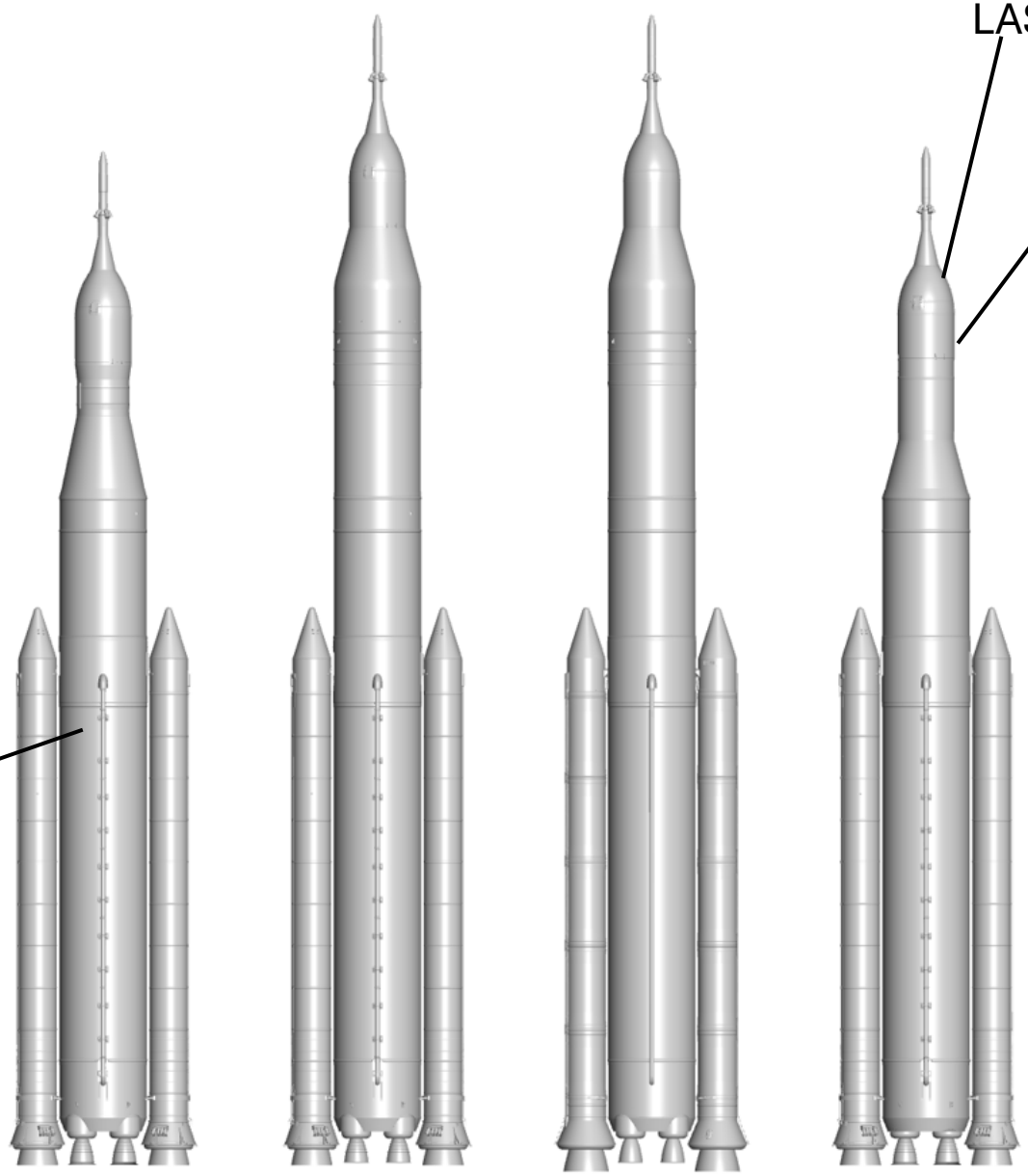
- Block 1
- Block 1B
- Block 2
- Block 1C (work in progress)



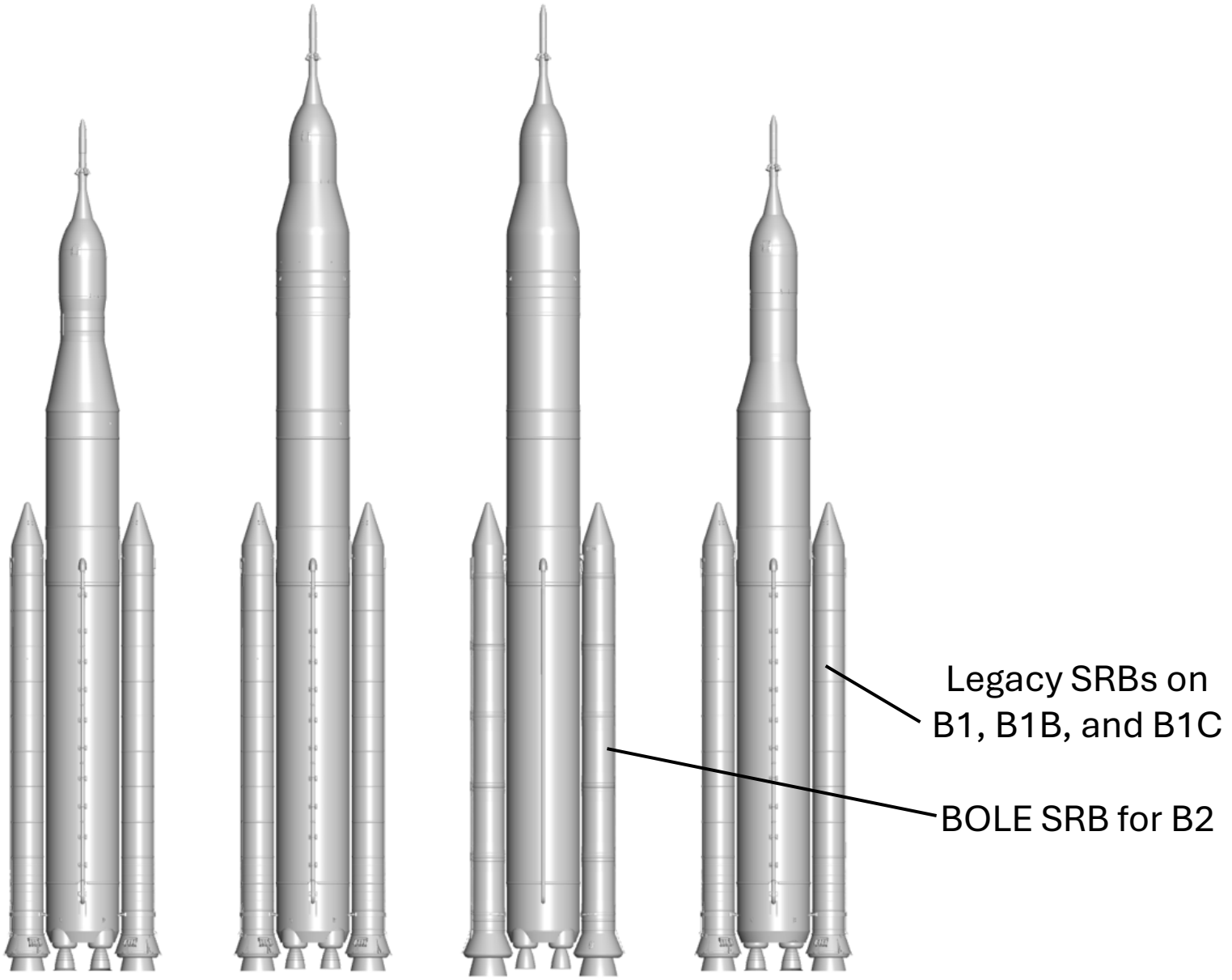
All are CREW vehicles
LAS and SM are the same

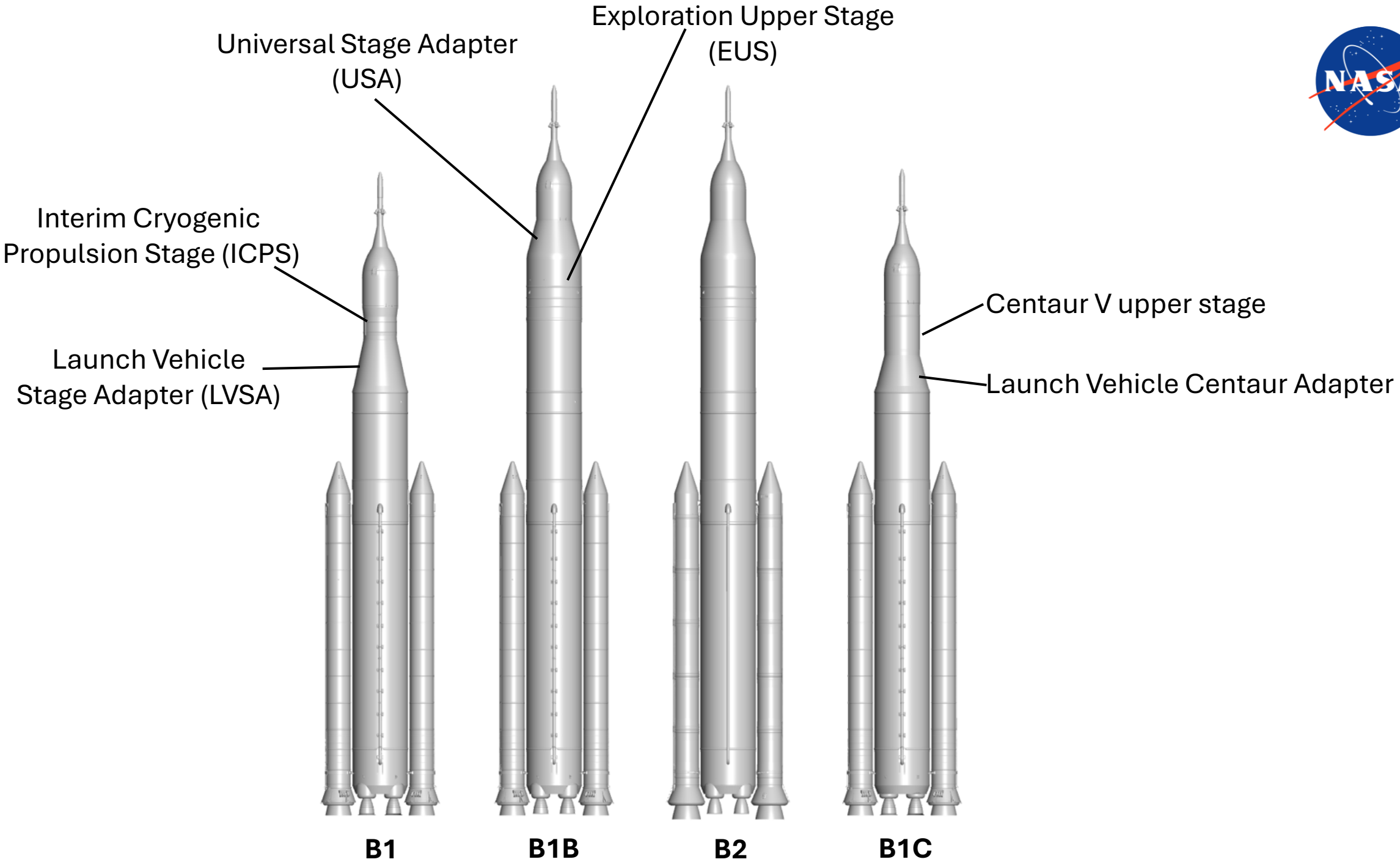
B1B and B2 have
same centerbody

Core stage same for all



B1 B1B B2 B1C

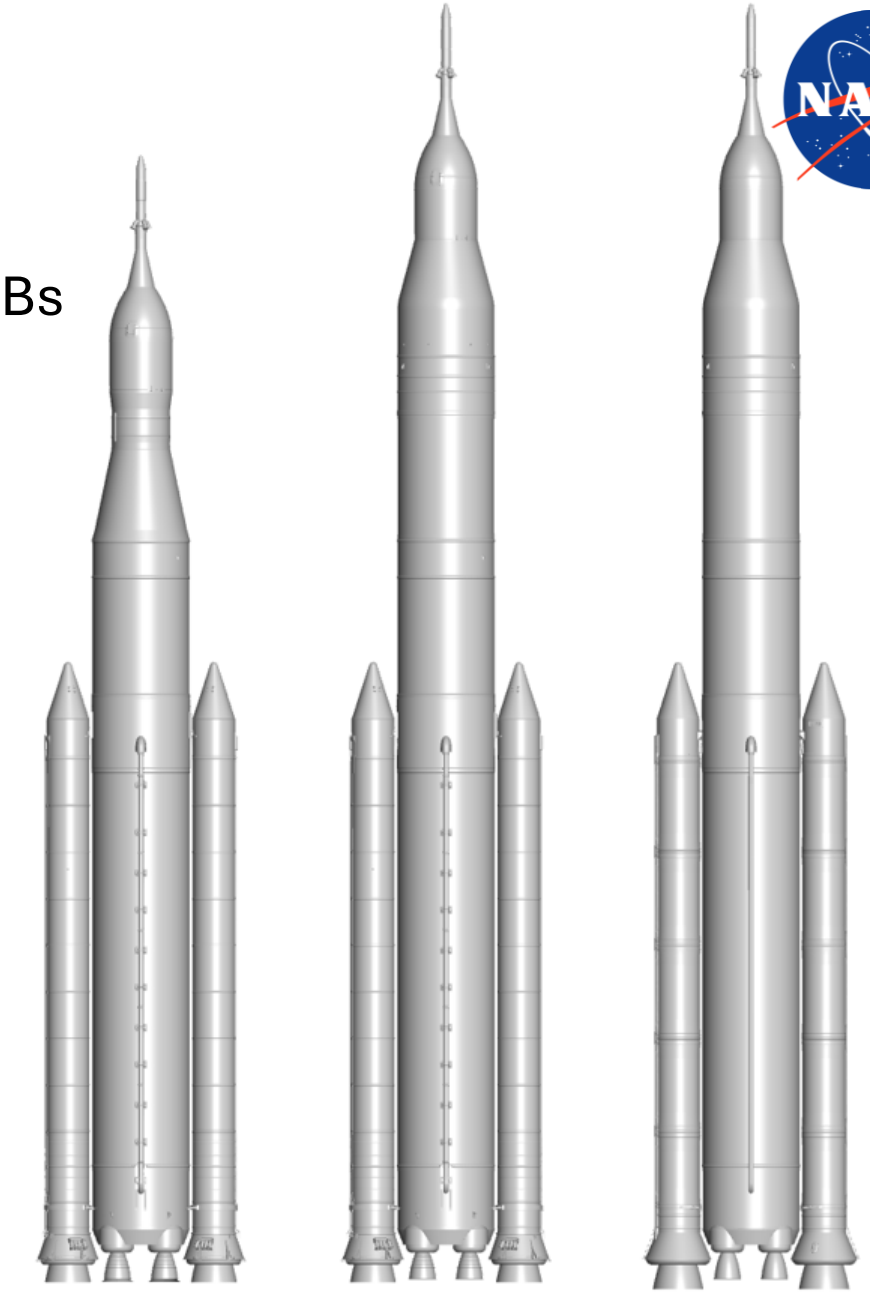

Legacy SRBs on
B1, B1B, and B1C
BOLE SRB for B2
B1
B1B
B2
B1C


Universal Stage Adapter (USA)
Exploration Upper Stage (EUS)
Interim Cryogenic Propulsion Stage (ICPS)
Launch Vehicle Stage Adapter (LVSA)
Centaur V upper stage
Launch Vehicle Centaur Adapter
B1
B1B
B2
B1C

Grid Sizes

- Ascent grids have all protuberances
  - Capped BSMs
- Booster separation has only protuberances that affect SRBs
  - BSM nozzle interiors

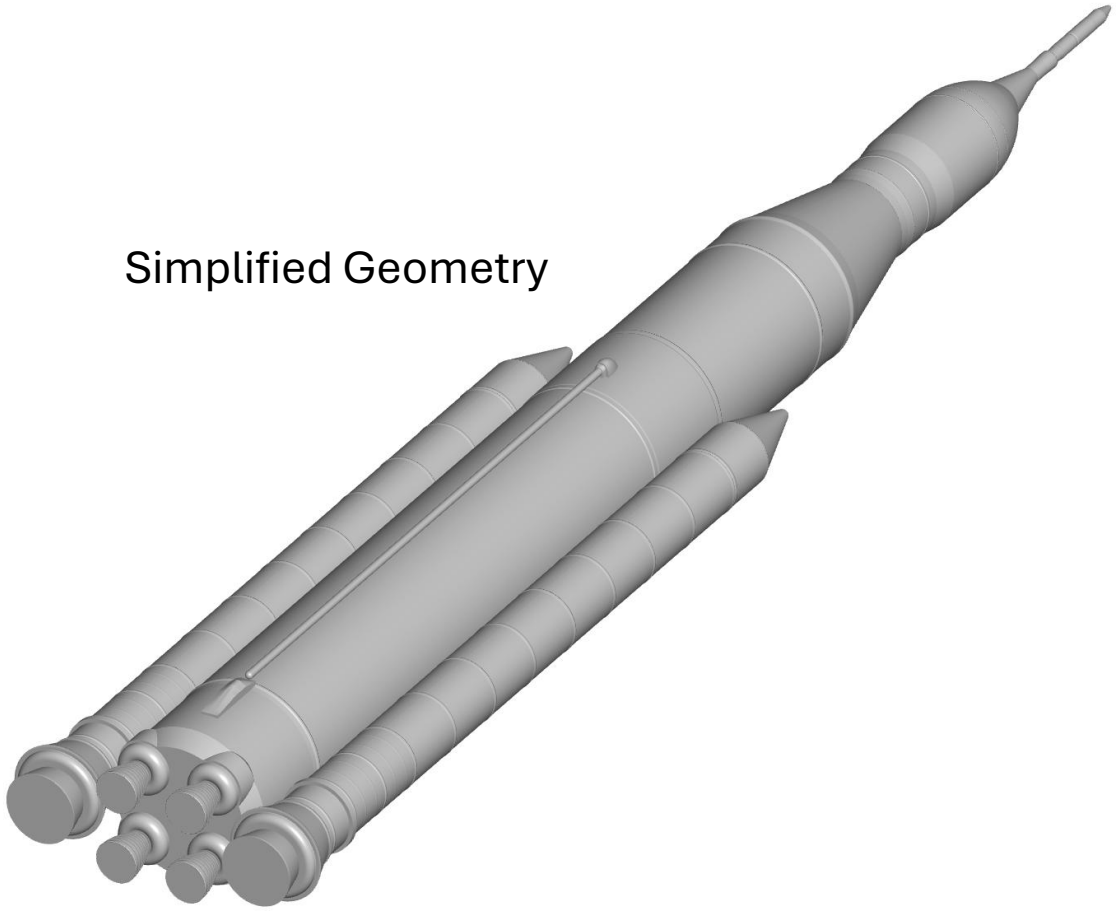
Mesh	NB Zones	NB Points	Post-AMR zones	Post-AMR Points
Block 1 Ascent	2263	793M	28000	1.1B
Block 1 Simple	82	49M	1300	70M
Block 1B Ascent	2314	788M	31000	1.1B
Block 2 Bsep	226	302M	11000	445M



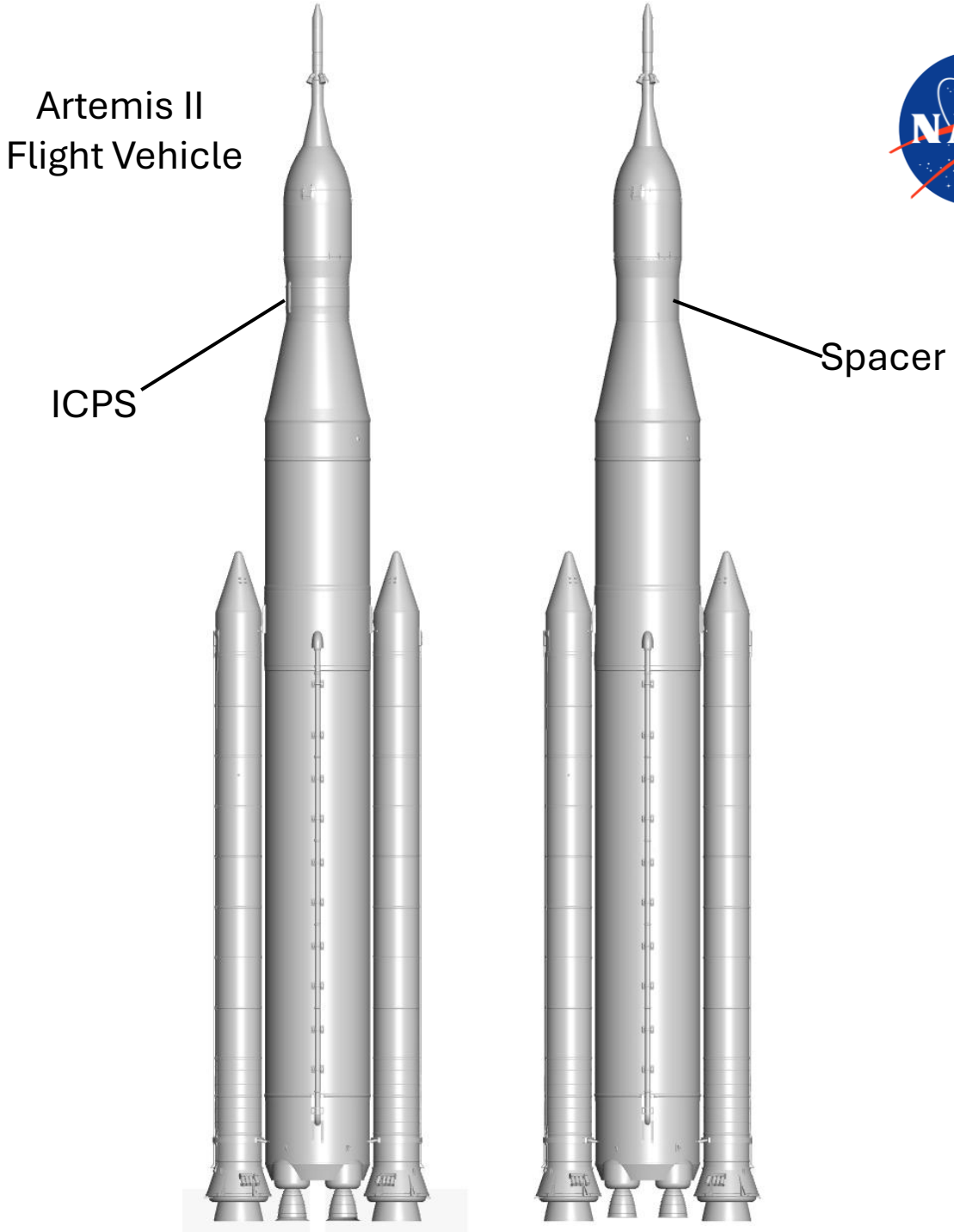
B1 **B1B** **B2**

Block 1

- Recent analyses completed
  - Artemis II ascent flight readiness
  - Artemis II off-nominal ascent (simplified geometry)
  - Artemis II post-flight ascent
  - Artemis III with ICPS and interim spacer

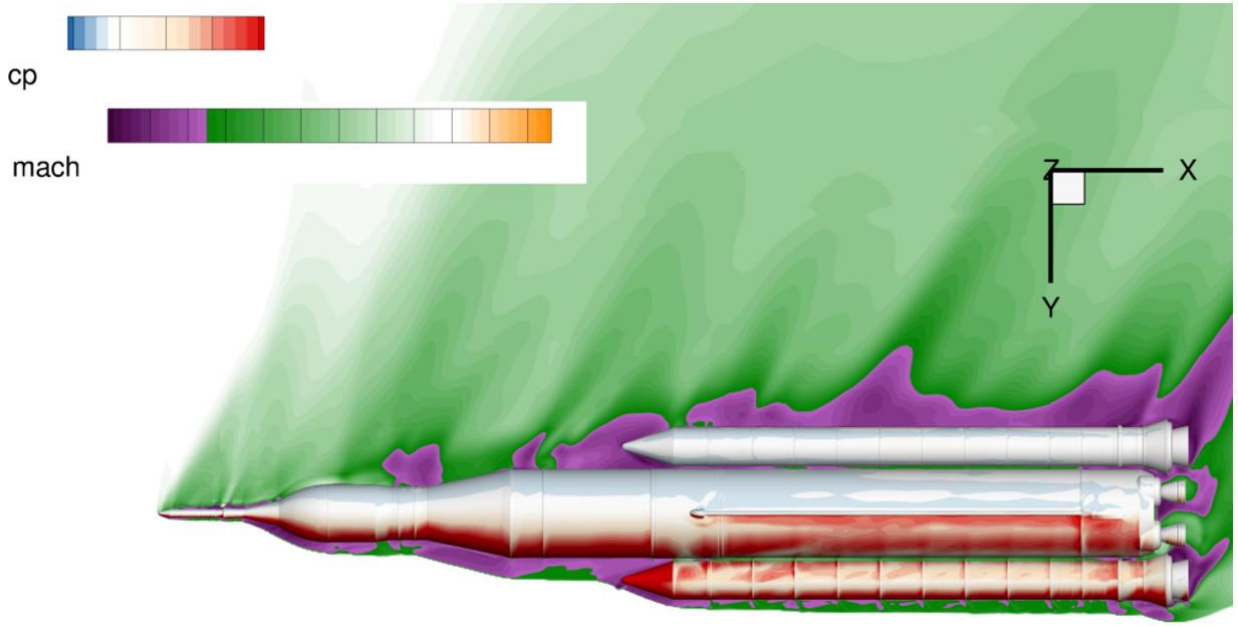


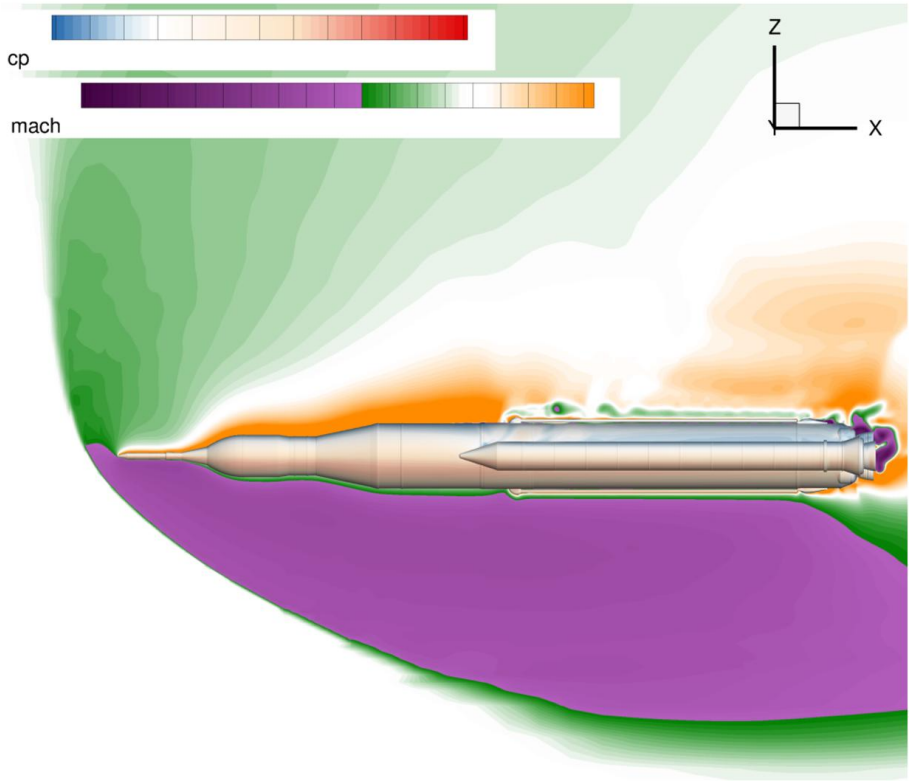
Simplified Geometry



Block 1- Off Nominal

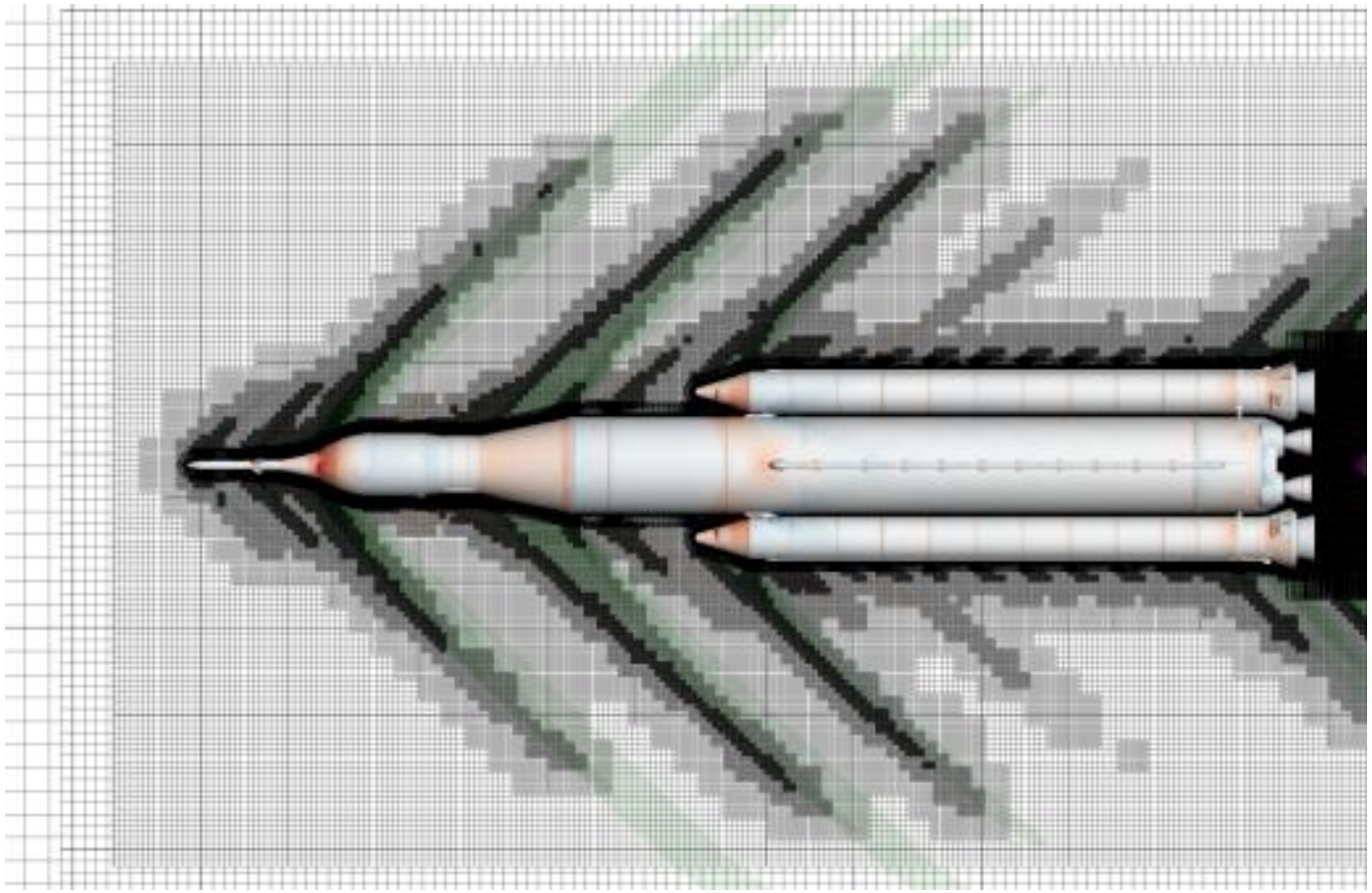
- Project required large number of off-nominal simulations, needed fast turn-around
  - Showed flexibility of overset methods and speed/robustness of OVERFLOW as a solver
  - Blended results from Cart3D (full geometry) and OVERFLOW (**simplified geometry**)

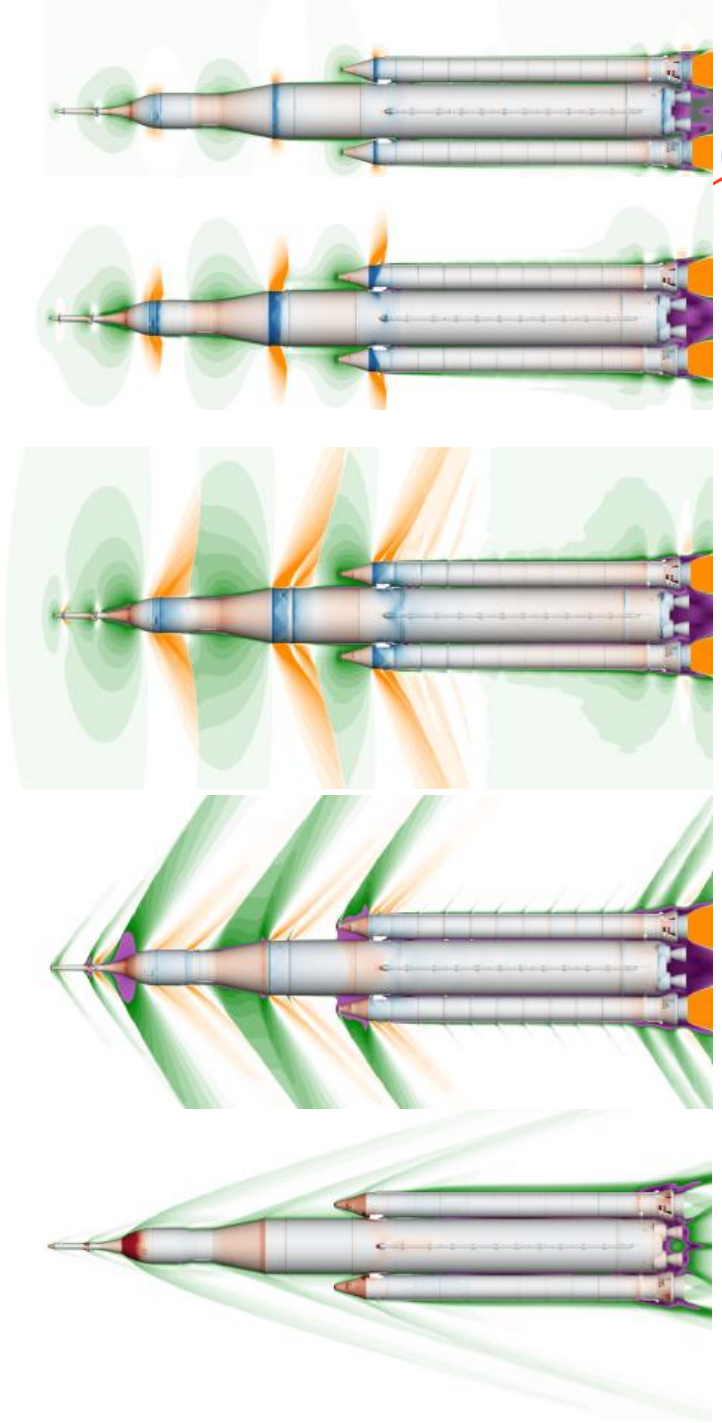




Block 1- Artemis II

- Artemis II flight readiness database
- Post-flight analysis along best estimated trajectory



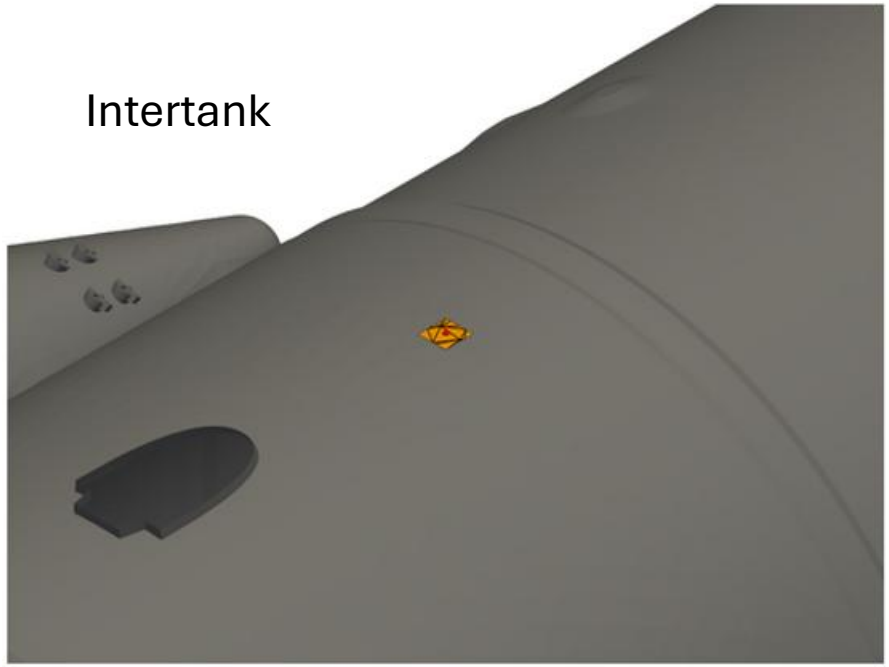


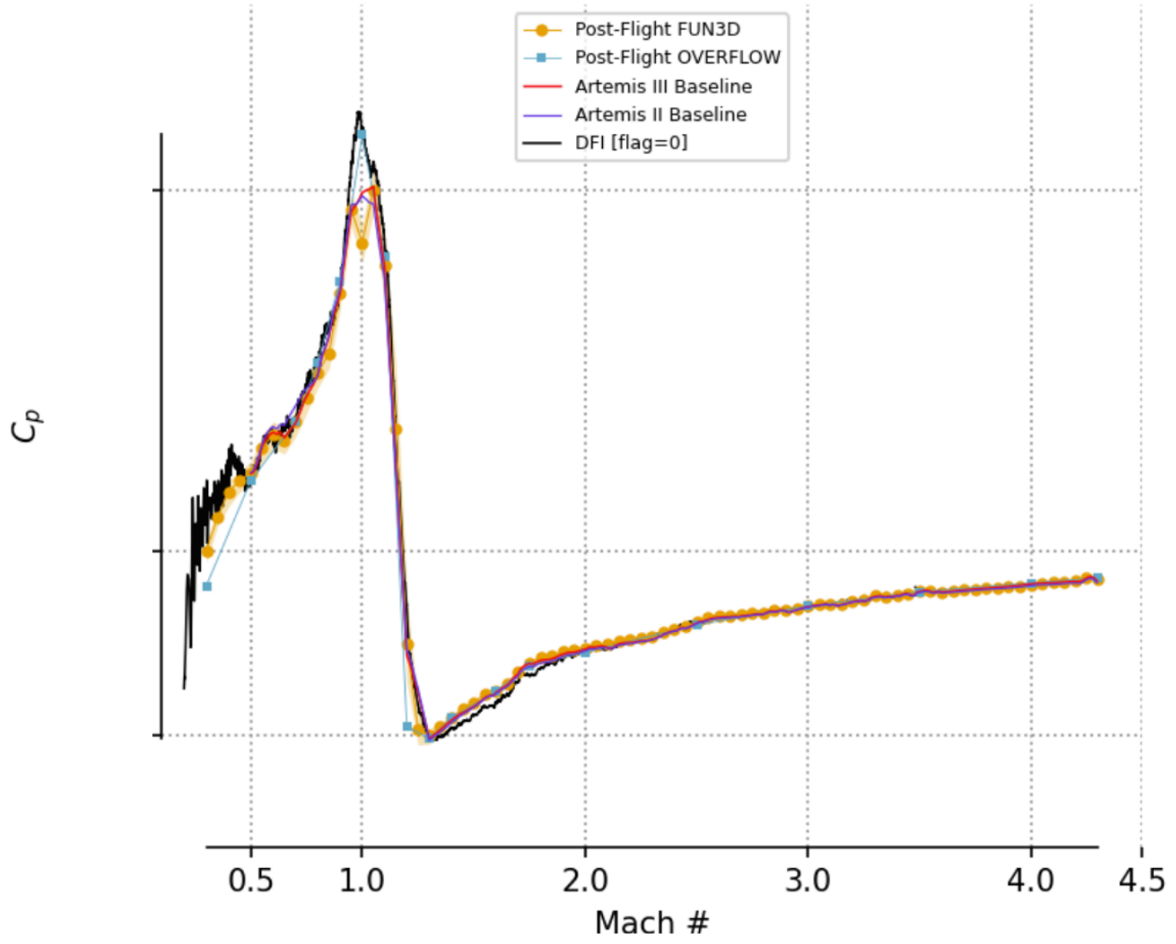
Artemis II Flight Data Comparison

- Looked at 157 pressure sensors (the ones that worked)
- CFD compared well to most sensors

DFI=Developmental Flight Instrumentation

Intertank



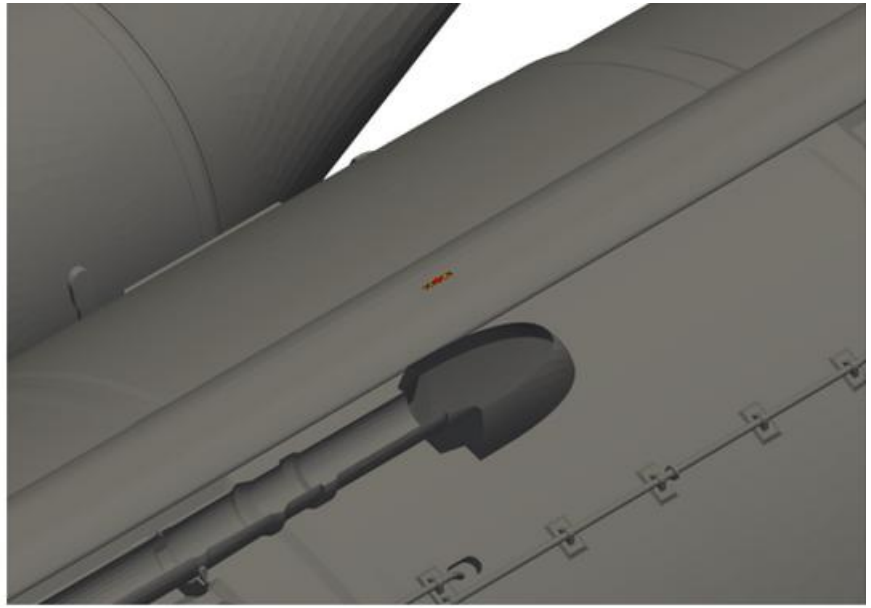


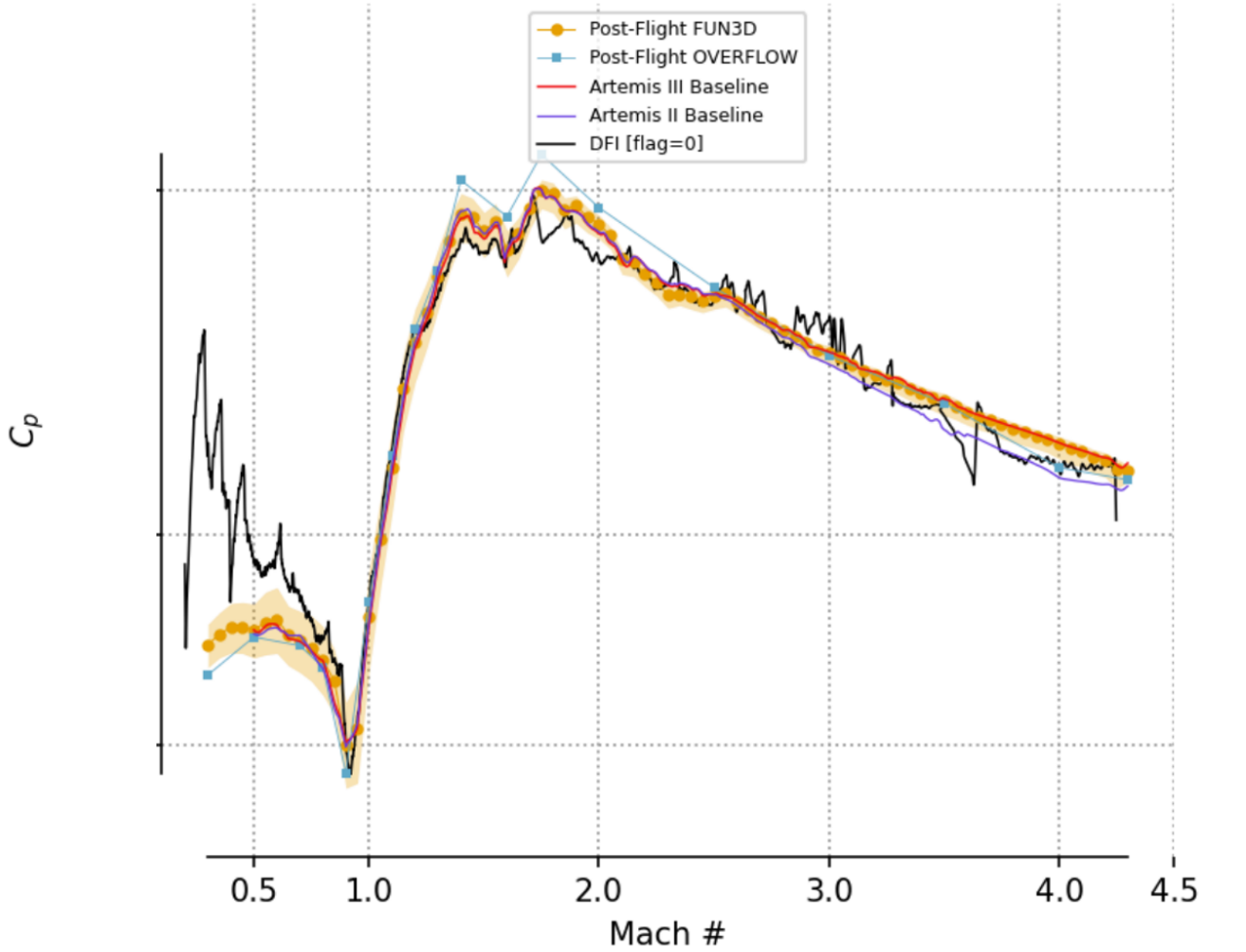
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DFI=Developmental Flight Instrumentation

Systems Tunnel





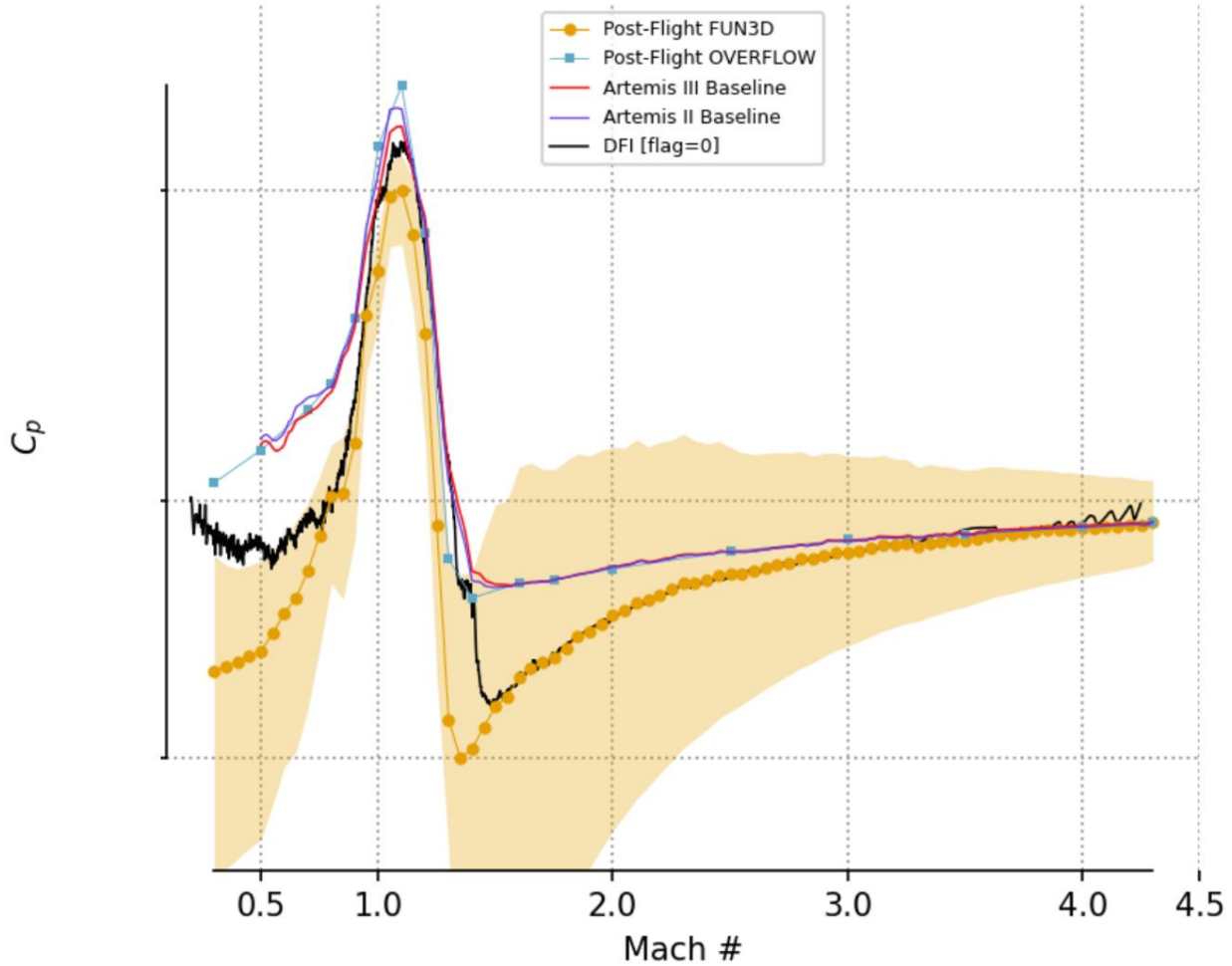
Artemis II Flight Data Comparison

- Sensor islands in the CFD mesh improved some comparisons

DFI=Developmental Flight Instrumentation

Sensor Island



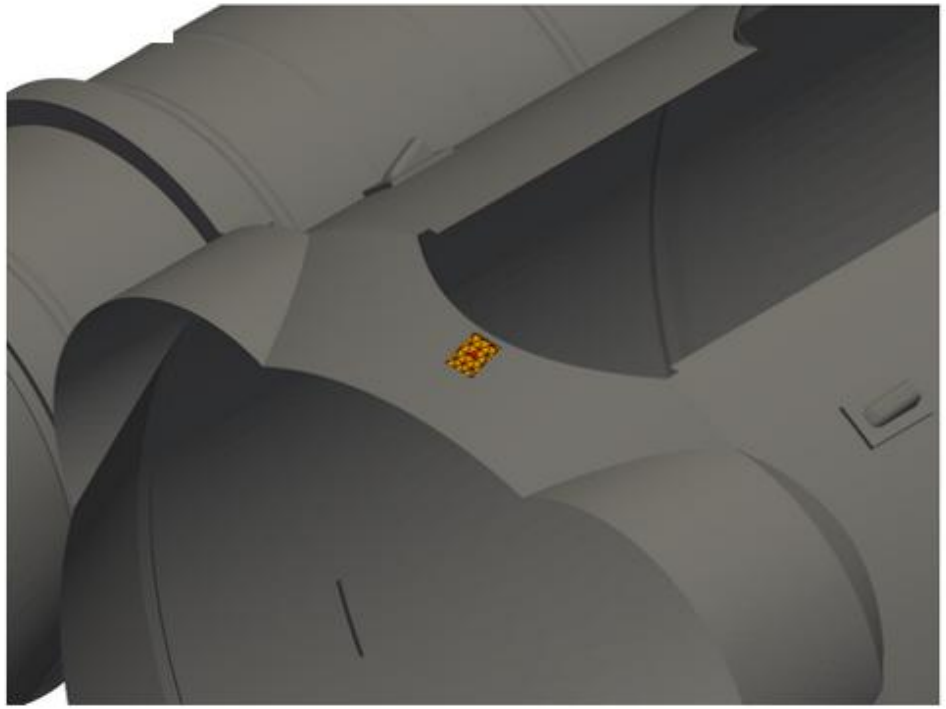


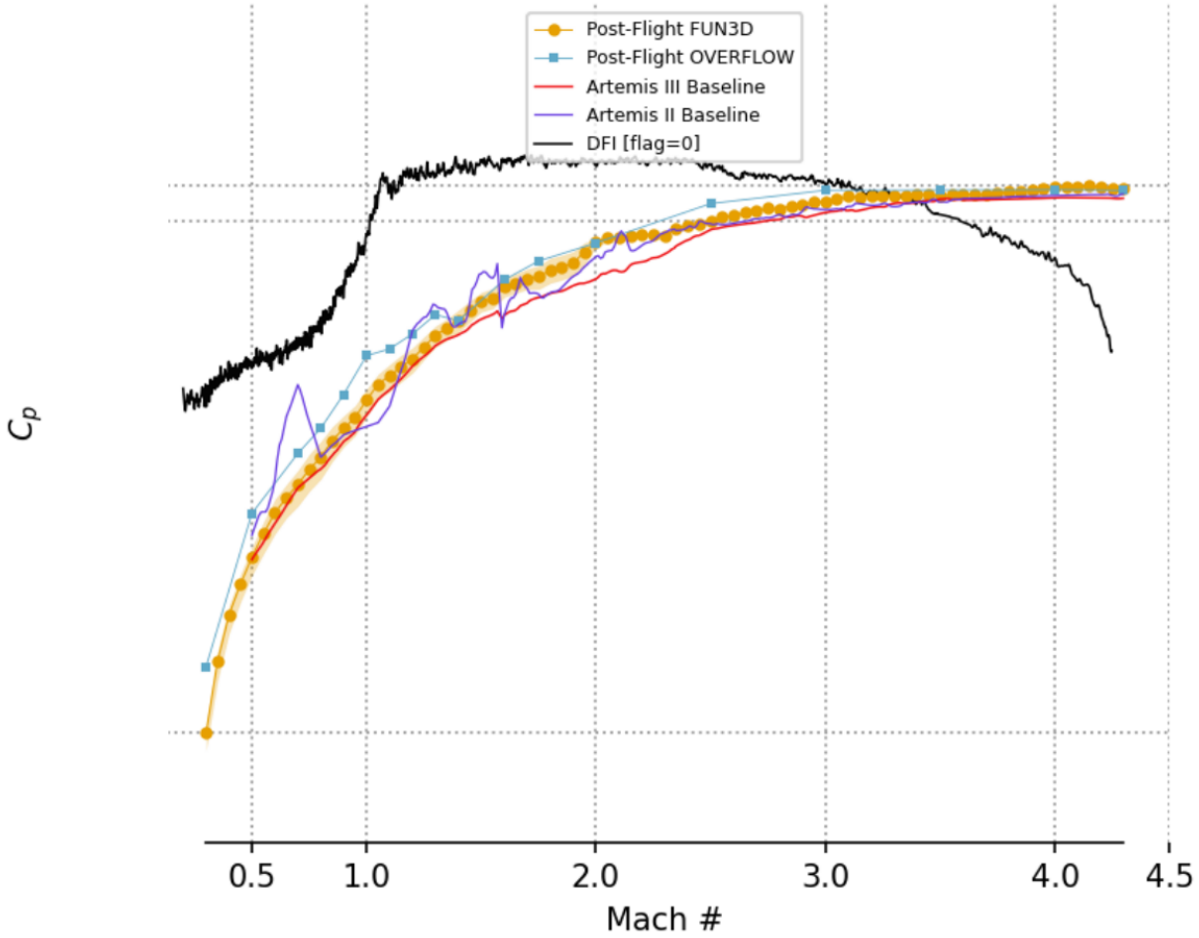
Artemis II Flight Data Comparison

- CFD did not compare well in all places

DFI=Developmental Flight Instrumentation

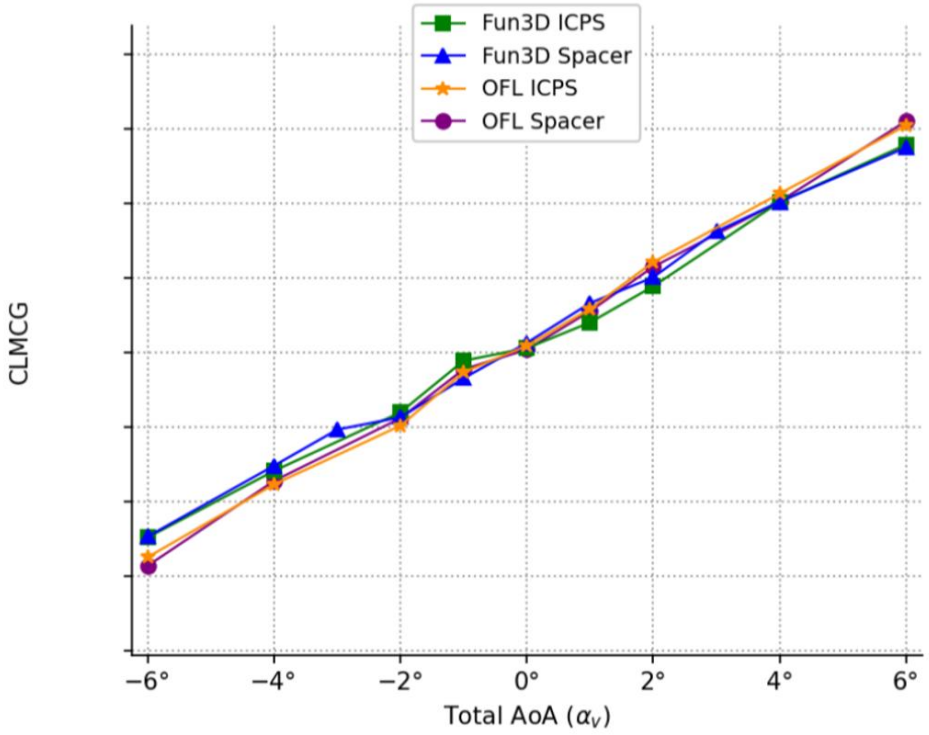
CORE base

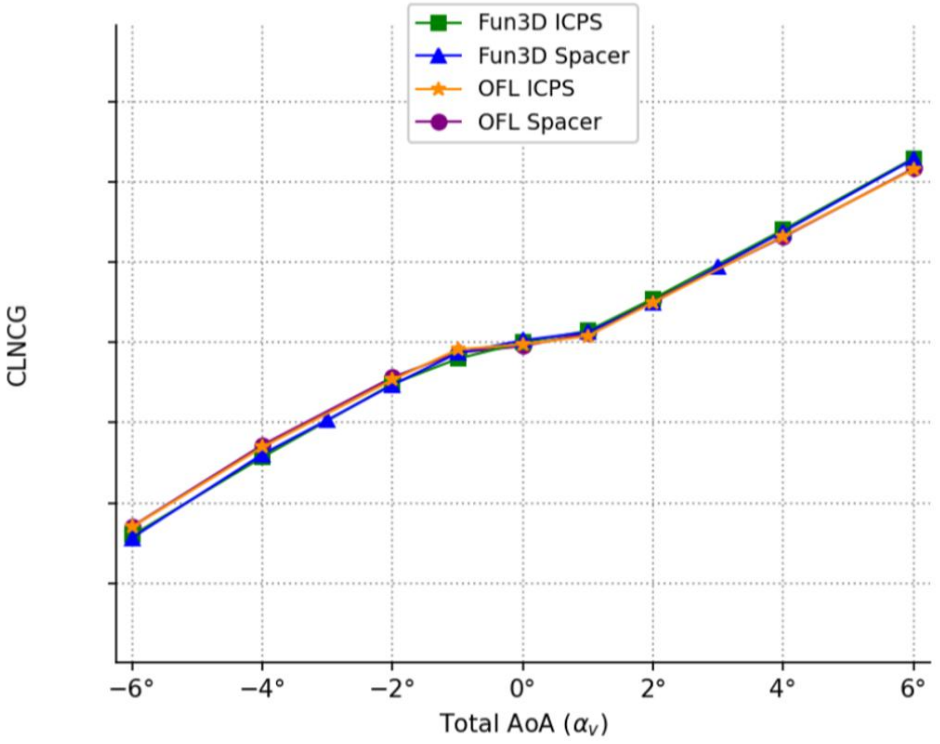


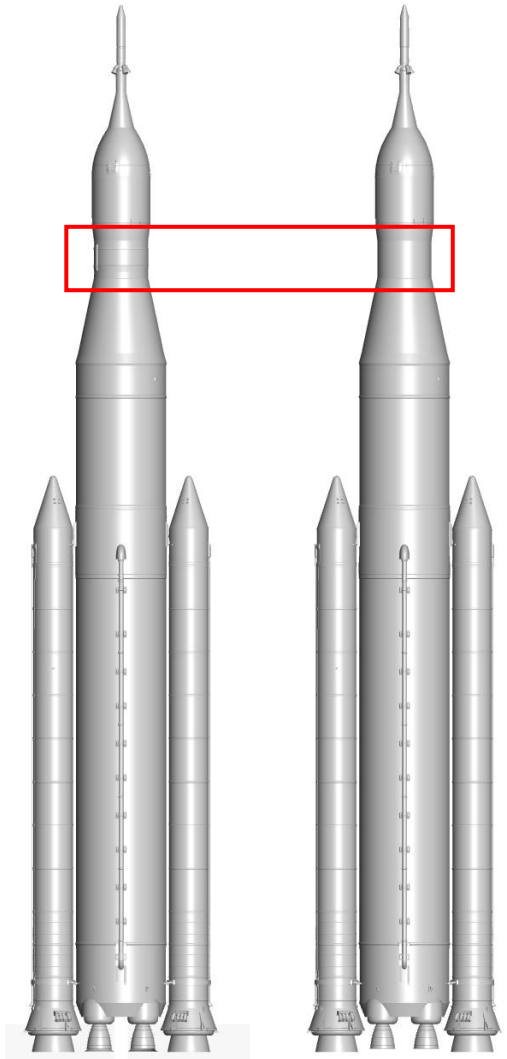


Pre-Artemis III ICPS vs. Spacer

- Overset grids make it easy to swap ICPS with spacer geometry
- Spacer was a temporary hold-over for new Artemis III upper stage
- Ran a set of check cases for a truncated centerbody FUN3D database
  - Checked FMs and lineloads for spacer influence

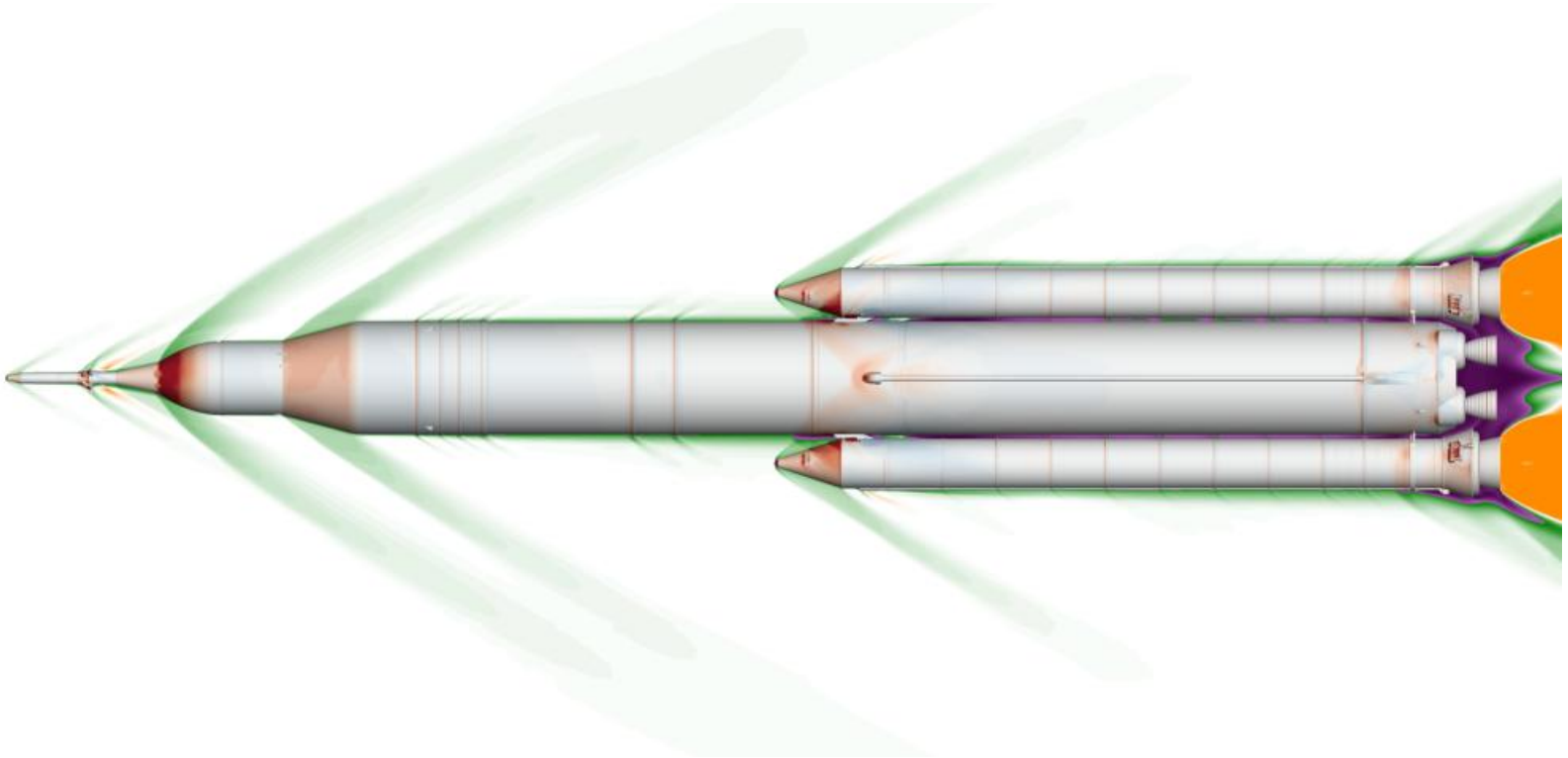


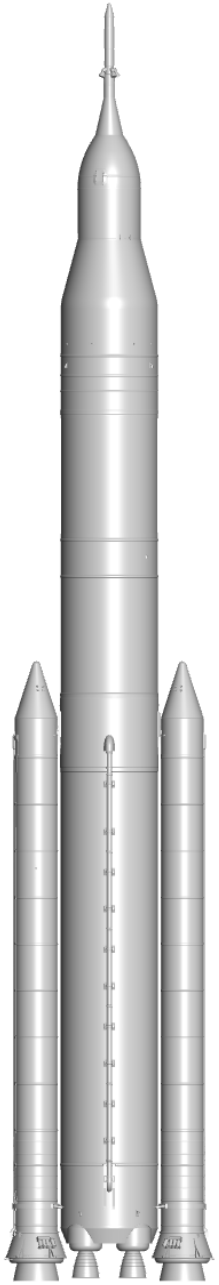




Block 1B

- Analyses completed
  - Ascent vehicle analysis cycle
  - Ascent vehicle analysis cycle with added protuberances including strakes
  - Debris Transport Analysis
- Block 1B work is frozen due to new SLS program direction



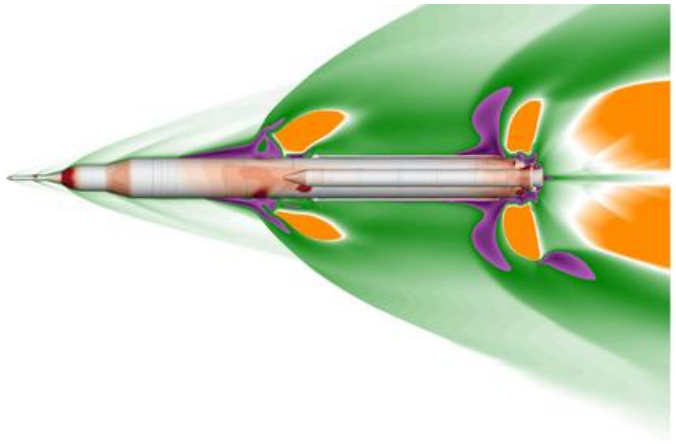


B1B

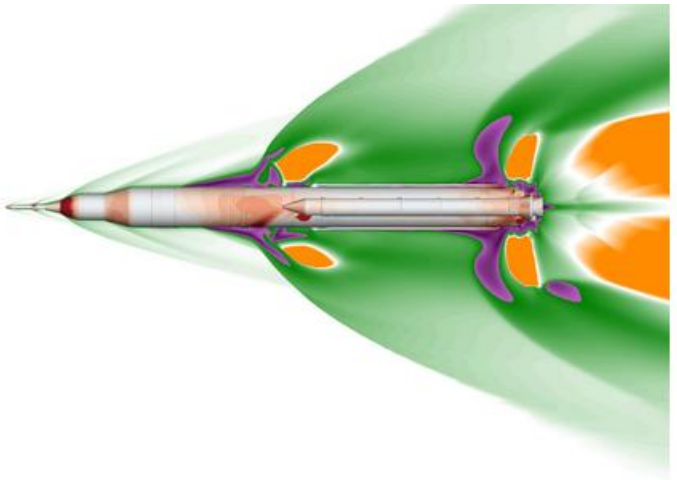
Block 2

- BOLE booster separation
  - Air vs multi-gamma
- BOLE boosters are not mirrored
- Differences between codes and nozzle gases wrapped into uncertainty

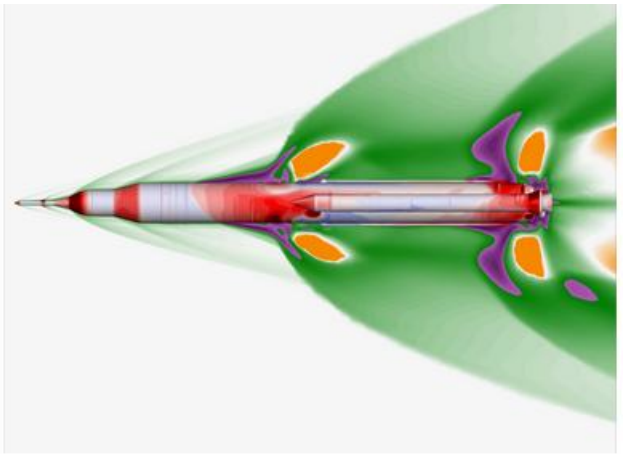




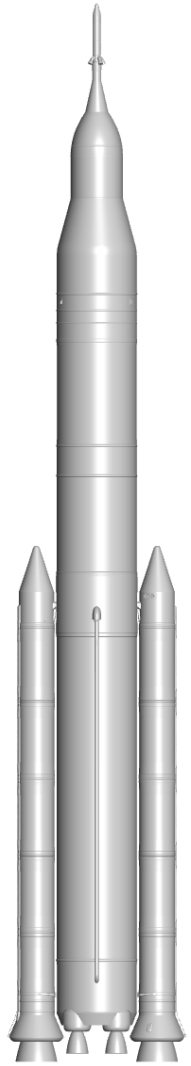
OVERFLOW multi-gamma CSEs and BSMs



OVERFLOW air CSEs and BSMs



FUN3D air CSEs and BSMs



Running on NAS

- Recent cases computed on either Rome or Turin nodes
- Turin nodes are 2-2.5x faster than Rome
  - OVERFLOW startup for Turin is much faster than Rome
- To run the billion-point cases
  - Pin rank0 to its own node
  - Use half the MPI processes as CPUs available for remaining nodes
- Some approximate computational times
  - Subsonic ascent cases 100k+ CPU hours (TUR)
  - Supersonic ascent cases ~20k CPU hours (TUR)
  - Booster separation cases ~50k CPU hours (ROM)



Future/Current Work

- Artemis III and beyond
- Standardizing SLS to increase launch cadence





Image credit: NASA

Acknowledgements

- Ames Computational Aerosciences Branch
- SLS Aerodynamics Task Team
- NASA Advanced Supercomputing
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- Sam Foxman



Image credit: NASA